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IN-VIVO MATHEMATICAL STUDY OF CO-INFECTION DYNAMICS OF HIV-1 AND *MYCOBACTERIUM TUBERCULOSIS*

GESHAM MAGOMBEDZE,[*,‡] WINSTON GARIRA[*,§]
and EDDIE MWENJE[†,¶]

Departments of [*]*Applied Mathematics and* [†]*Applied Biology*
National University of Science and Technology
P. O. Box AC939 Ascot, Bulawayo, Zimbabwe
[‡]*gmagombedze@nust.ac.zw*
[§]*wgarira@nust.ac.zw*
[¶]*emwenje@nust.ac.zw*

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Human Immunodeficiency Virus type-1 (HIV-1) fuels the pathogenesis of *Mycobacterium tuberculosis* (Mtb) in humans. We develop a mathematical model in an attempt to understand the immune mechanisms that are involved during the co-infection of Mtb and HIV-1. Our study reveals that infection of an Mtb infected individual with HIV-1 results in fast development of active TB. The mathematical model analysis and simulations show that Mtb infection is linked to HIV infection through macrophages and CD4+ T cells. The study shows that depletion of macrophages and CD4+ T cells by HIV-1 worsens the picture of Mtb infection and in-turn Mtb infection affects the progression of HIV-1 infection since it is also capable of inducing rapid replication of HIV. Our analytical and numerical simulations show that macrophages are a potential reservoir of HIV particles during HIV-1 infection. Co-infection simulations reveal that co-infection exacerbates more the pathogen that caused the first infection. Simulations also show that co-infection disease progression patterns converge to a similar trend after a considerable time interval irrespective of which pathogen first caused infection and the second pathogen that caused co-infection. This work suggests directions for further studies and potential treatment strategies.

Keywords: Human Immunodeficiency Virus (HIV); *Mycobacterium tuberculosis* (Mtb); CD4+ Helper T Cells; Co-Infection; Mathematical Modeling.

1. Introduction

Since 1985 there has been a renewed epidemic of tuberculosis (TB) that was previously thought to be in check.[1] Today, it remains the leading cause of death by pathogen induced disease world wide. It is believed that this recent increased incidence is in large part due to HIV. This study focuses mainly on the effect that HIV-1 has on Mtb infection and vice-versa. First, Mtb infection induces cytokine expression which result in an increase of transcriptional activity of HIV.[2,3] Pro- and anti-inflammatory cytokines are produced during Mtb infection, which include IL-2, IL-4, IL-6, IL-10, IL-12, IFN-γ, and TNF-α. High levels of TNF-α are found at infection sites in pulmonary TB. TNF-α has been found to be associated with high HIV replication in TB co-infected individuals.[2–4] Second, HIV infected individuals are at an increased risk of developing TB in the active form.[5] Third, TB is the most common HIV related complication world wide. Fourth, HIV infected individuals are not only at a greater risk of acquiring TB, but reactivation of latent TB is greatly increased due to the fact that the very cells that hold the latent TB in check (the CD4+ T lymphocytes) are precisely the cells that are rendered dysfunctional in HIV infected individuals.[1,6]

HIV-1 belongs to the lentivirus subfamily of retroviruses. As with all retroviruses, the viral genes in infectious particles are carried as RNA but upon infection of the host cell, reverse transcriptase catalyses the synthesis of double stranded DNA viral genome.[6] The major hallmarks of HIV infection include the destruction of helper CD4+ T lymphocytes and subsequent loss of immune competence.[6,7] HIV causes loss of CD4+ T cells by directly infecting and killing them, and inducing programmed cell death (apoptosis) via the Fas–FasL pathway.[8] It impairs cell function, because of an aberrant reaction to the infection.[7] Once the CD4+ T cells are depleted, this results in a weakened immune system. The evolution of HIV has been characterized into four main stages of disease progression. First is the initial innoculum, when the virus is introduced into the body. Second, is the initial transient, a relatively short period of time when the T cell population and virus population are in great flux.[5,6] This is followed by the third stage, clinical latency. This is a period of time when there are extremely large numbers of virus and T cells undergoing incredible dynamics, the overall result of which is an appearance of latency (disease steady state). Finally there is AIDS. This is characterized by the T cell dropping to very low numbers and uncontrolled viral growth, resulting in death.[5] This is also characterised by development of opportunistic infections and other pathological conditions. Strong HIV-specific CD4+ T cell responses and strong HIV-specific CD8+ T cell responses during acute HIV-1 infection are associated with good control of early HIV-1 replication.

On the other hand, Mtb is a slowly growing, facultative intracellular pathogen that can survive and multiply inside macrophages and other mammalian cells. The bacilli spread from the site of initial infection in the lungs through the lymphatics or blood to other parts of the body.[9–11] The immune response following the first exposure to Mtb is multifaceted and complex. It usually develops in the alveoli of the lungs and the alveolar macrophages are an ideal target for Mtb. The bacteria multiply inside the macrophages up to a threshold limit after which the macrophages burst, releasing more bacteria.[12] The bacteria prefers intracellular growth as opposed to extracellular growth. In their resting state, not only are alveolar macrophages poor at destroying the bacteria, but Mtb can also inhibit their ability to kill phagocytized bacteria, most likely by preventing phagosome–lysosome fusion.[9–11,13,14] If the macrophage fails to receive sufficient stimulation for activation, it is unable to clear its' resident bacteria. These chronically infected macrophages eventually either die due to a large number of resident bacteria or are destroyed by a CTL response.[9–15] After encountering the Mtb bacteria, resting macrophages release cytokines which trigger activation of the macrophages. Immunologic control of Mtb infection is based on a Th1 response. IL-12 is induced following phagocytosis of Mtb which drives development of Th1 response.[9–11,13,14] IFN-γ is another central cytokine in the control of Mtb. This cytokine is produced by both CD4+ T cells and CD8+ T cells and is important in macrophage activation.[9–11,13,14] T cells are responsible for killing infected macrophages that are unable to destroy their resident bacteria. This is accomplished via a Fas–Fas ligand apoptotic pathway by CD4+ T cells[9–11,13,14,16] and via other cytotoxic mechanisms such as granule and perforins produced by CD8+ T cells and possibly CD4+ T cells.[10,11,13,16,17] CD4+ T cells are involved in recognition of antigens that have been processed in phagosomes and presented as small peptide fragments in the context of MHC class II molecules on the surface of antigen presenting cells such as monocytes macrophages, or dendritic cells. CD8+ T cells, on the other hand, recognize antigens that have been processed in the cytosol and that are presented in the context of MHC class I molecules on the cell surface.[9–11,13,14] In general CD4+ T cells help to amplify the host immune response by activating effector cells and recruiting additional immune cells to the site of disease, whereas CD8+ T cells are more likely to be directly cytotoxic to target cells.

In this paper, we focus on investigating how HIV affects the course of Mtb infection and in-turn how Mtb infection affect the progression of HIV infection. This work seeks to address how HIV and Mtb infections are linked and to shade light on the human immune mechanisms involved. Kirschner was the first to present a model[1] of Mtb and HIV dynamics in the lymph tissue. However, the work by Kirschner[1] did not separate CD4+ T cell dynamics from CD8+ T cells, and also did not consider the dynamics of Mtb and HIV specific CTLs. This work considers the infection dynamics that occur in the lungs during TB and HIV co-infection. There is little or no infection of macrophages that is noticed in the lungs in the absence of co-infection,[18–20] but during TB/HIV co-infection macrophages are significantly infected by HIV. This work helps to determine how HIV aggravates the course of TB in the lungs since TB is a disease of the lungs and inturn how TB also accelerates the progression of HIV.

This paper is organized as follows: Section 2 present mathematical models for Mtb and HIV separate infections, their description, and their mathematical analysis. The models describe the immune response mechanisms of the human body against Mtb infection and the HIV infection, respectively. In Sec. 3, we present a mathematical model for Mtb/HIV co-infection with its description and mathematical analysis. Section 4 includes numerical simulations and results. Discussion of results is done in Sec. 5.

2. Mathematical Modeling

The concepts introduced above are incorporated into a model to describe the immune system of bacteria virus interactions. These interactions are taken to be from the lung tissue. We use the lung compartment because the T lymphocyte distribution in branchoaveolar lavage fluid (BAL) is to a lesser extent a reflection of the T lymphocyte distribution in the peripheral blood that is, 65% to 75% are CD3+, 40% to 45% are CD4+, and 20% to 25% are CD8+ T cells.[21] Nevertheless, a model encompassing separate compartments that is the blood and lung would produce more elaborate results. The model outlines interactions among three macrophage populations (resting macrophages (M_r), Mtb infected macrophages (M_{ib}), and HIV infected macrophages (M_{iv})), one bacterial population (T_b), two populations of CD4+ T cells, that is, infected (T^*) and uninfected (T) with the virus, two populations of CTLs (HIV specific CTLs (C_{hv}), and Mtb specific CTLs (C_b)) and the viral population (V). The first model, describes the dynamics of Mtb infection and the immune system. The second model, models the dynamics of HIV infection and the immune system. The co-infection dynamics of Mtb/HIV interactions with the immune system are considered in the third model.

2.1. *Mtb model without HIV infection*

This model describes the interactions of Mtb pathogen and the immune system. The model presented here is a simplification of our earlier model.[22]

$$\frac{dM_r}{dt} = s_m + p_m T_b M_r - k_i T_b M_r - \mu_m M_r \tag{1}$$

$$\frac{dM_{ib}}{dt} = k_i T_b M_r - k_b M_{ib} - k_a \left(\frac{M_{ib}}{M_{ib} + A}\right) T - k_l M_{ib} C_b - \mu_{mi} M_{ib} \tag{2}$$

$$\frac{dT_b}{dt} = k_b N_b M_{ib} - \gamma_1 T_b M_r - \gamma_2 T_b C_b + r_m T_b + k_l N_c M_{ib} C_b \tag{3}$$

$$\frac{dT}{dt} = s_t + r_1 \left(\frac{T T_b}{T_b + B}\right) - \mu_t T \tag{4}$$

$$\frac{dC_b}{dt} = s_b + p_b \left(\frac{M_{ib}}{M_{ib} + R_b}\right) T C_b - \mu_{cb} C_b \tag{5}$$

Equation (1) models the dynamics of resting macrophages (M_r). Here s_m is the source of macrophages from bone marrow progenitor monocyte differentiation.[12,22] New macrophages are recruited in response to Mtb infection at rate p_m. k_i represents the rate at which Mtb (T_b) infects resting macrophages. The last term represents the natural death of resting macrophages.

Equation (2) represents the dynamics of Mtb infected macrophages. The term $k_i T_b M_r$ represents the source of Mtb infected macrophages from term 3 of Eq. (1). The second term represents the bursting of Mtb infected macrophages at rate k_b as the bacterial burden increases.[10,13,14,22] The third term represents the apoptotic death of Mtb infected macrophages by T cells through the Fas–Fas apoptotic pathway. Lytic killing by Mtb specific[10,11,13,14] CTLs (C_b) is modeled by the fourth term.[13,16,17,22] The last term represents the natural death of Mtb infected macrophages at rate μ_{mi}. In this model, we associate apoptotic death of Mtb infected macrophages due to CD4+ T cells with the death of the intracellular bacteria. We consider only lytic death to be associated with release of intracellular bacteria (we assume that the amount of bacteria released through apoptosis is insignificant compared to the amount released through lytic killing since $k_a(\frac{M_{ib}}{M_{ib}+A}) < k_l Mib$ and $k_a < k_l$).

Equation (3) models the dynamics of free bacterial particles. The first term represents the source of bacteria from the burst term of (2), where N_b is the burst size.[12,22] Second and third terms represent the loss of bacteria through killing by macrophages and by Mtb specific CTLs.[16,17,22] The fourth term represents the bacterial multiplication at rate r_m. The last term represents the number of bacterial particles that are released into the extracellular environment when infected macrophages are lysed by Mtb specific CTLs at rate k_l.[16,17,22]

Equation (4) models the dynamics of health CD4+ T cells. s_t is the source of CD4+ T cells from the thymus.[5] The second term represents the proliferation of CD4+ T cells due to Mtb infection at proliferative rate r_1. Priming and proliferation of CD4+ T cells is proportional to the density and efficiency (maturation, trafficking, and antigen presentation) of antigen presenting cells and these activities are dependent on the density of the pathogen $(\frac{T_b}{T_b+B})$ at the site of infection. Antigen detection, infected and activated macrophages and a combination of these mechanisms creates a cytokine and chemokine environment that drive the further priming of T cells.[9,10,13,14] The last term represents the natural death of CD4+ T cells at rate μ_t.

Equation (5) models the dynamics of Mtb specific CTLs (C_b). The term s_b represents the source of CD8+ T cells from the thymus. p_b is the proliferative rate of Mtb specific CTLs. In Eq. (4) we used the density of the pathogen to model the proliferation of CD4+ T cells. Here we use the density of infected macrophages for the proliferation of Mtb specific CTLs since they secret cytokines that derive futher priming of Mtb CTLs. The effects of the pathogen (T_b) density are also indirectly captured through T (CD4+ T cells that depend on the density of the Mtb pathogen at the site of infection) in this term. μ_{cb} represents the natural death rate of Mtb specific CTLs.

2.1.1. *Model analysis*

The equilibria of this model are attained by setting the right-hand side of Eqs. (1)–(5) to zero. It follows then that the system of Eqs. (1)–(5) has a disease-free equilibrium

$$E_{00} = (\hat{M}_r, \hat{M}_{ib}, \hat{T}_b, \hat{T}, \hat{C}_b) = \left(\frac{s_m}{\mu_m}, 0, 0, \frac{s_t}{\mu_t}, \frac{s_b}{\mu_{cb}}\right).$$

The model also has an endemic equilibrium given by

$$E_{01} = (\bar{M}_r, \bar{M}_{ib}, \bar{T}_b, \bar{T}, \bar{C}_b),$$

where $\bar{M}_r$, $\bar{M}_{ib}$, $\bar{T}_b$, $\bar{T}$, and $\bar{C}_b$ are given by expressions (6)–(10), respectively. The endemically infected state represent two possible disease outcomes that is TB latency and TB active disease. TB latency occurs when the human immune response is not strong enough to eliminate the infection but relatively strong to prevent further progression of infection to cause active disease. Weakening of these immune mechanisms result in the activation of latent TB to active TB or further progression of infection to cause active disease.

The equilibrium value of resting macrophages is given by

$$\bar{M}_r = \frac{s_m}{(\mu_m + \bar{T}_b(k_i - p_m))}. \tag{6}$$

Expression (6) shows that the equilibrium value of resting macrophages depend on Mtb pathogen. Increasing the recruitment rate (p_m) of resting macrophages to a value greater than the infection rate (k_i) helps to control the infection progression. The population of Mtb infected macrophages is given by

$$\bar{M}_{ib} = \frac{-w_0 + \sqrt{w_0^2 - 4w_1w_2}}{2w_1}, \tag{7}$$

where

$$w_0 = k_bA + A(k_l\bar{C}_b + \mu_{mi}) + k_a\bar{T} - k_iT_bM_r,$$
$$w_1 = k_b + k_l\bar{C}_b + \mu_{mi}, \quad w_2 = -k_i\bar{T}_b\bar{M}_rA.$$

All cell variables assume values greater or equal to zero for the model to be biologically meaningful. $w_0 \geq 0$ (if $k_bA + A(k_l\bar{C}_b + \mu_{mi}) + k_a\bar{T} > k_iT_bM_r$), $w_1 \geq 0$ and $w_2 < 0$ since all model parameters assume positive values therefore, Eq. (7) is positive since $-w_0 + \sqrt{w_0^2 - 4w_1w_2} \geq 0$ what ever the value of w_0 is. We take the positive value of M_{ib} as the feasible value of Mtb infected macrophages at the disease equilibrium state. This expression shows that the equilibrium value of Mtb infected macrophages depend on Mtb specific CTLs and resting macrophages. Increasing w_1 (burst size and CTL killing of infected macrophages) reduce the level of infected macrophages. The steady-state value of the Mtb pathogen is given by

$$\bar{T}_b = \left(\frac{k_bN_b\bar{M}_{ib} + k_lN_c\bar{M}_{ib}\bar{C}_b}{\gamma_1\bar{M}_r + \gamma_2\bar{C}_b - r_m}\right). \tag{8}$$

We deduce from expression (8) that increasing the burst size, burst rate, and rate of bacterial multiplication will increase the bacterial burden, while killing or elimination of bacteria by Mtb specific CTLs and resting macrophages will reduce the bacterial burden. The equilibrium value of health CD4+ T cells is given by

$$\bar{T} = \left(\frac{s_t}{\mu_t - r_1\frac{\bar{T}_b}{\bar{T}_b + B}}\right). \tag{9}$$

This shows that CD4+ T cells increase in population due to the presence of the Mtb pathogen. This follows that increasing mechanisms that increase proliferation of T cells in Mtb infection help to increase levels of T cells, hence increase in the performance of the immune cells. The population density of Mtb specific CTLs is given by

$$\bar{C}_b = \left(\frac{s_b}{\mu_{cb} - p_b \left(\frac{\bar{M}_{ib}}{\bar{M}_{ib} + R_b} \right) \bar{T}} \right). \tag{10}$$

We conclude from expression (10) that there exists an inverse relationship between the density of Mtb infected macrophages and CD4+ T helper cells, that is, for $\bar{C}_b$ to be positive, $\mu_{cb} > p_b \bar{T}(\frac{\bar{M}_{ib}}{\bar{M}_{ib}+R_b})$, $(\frac{\bar{M}_{ib}}{\bar{M}_{ib}+R_b}) < (\frac{\mu_{cb}}{p_b \bar{T}})$, $\Rightarrow (\frac{\bar{M}_{ib}}{\bar{M}_{ib}+R_b}) \propto \frac{1}{\bar{T}})$. This analysis shows that increasing the level of infected macrophages leads to the production of inflammation inducing cytokines and chemokines that reduce the levels of Th1-type of T cells. This might be linked to the switching from the Th1 type of immune system to the Th2 type that is associated with active disease.[10,11,13,23]

2.1.2. *Reproduction ratio*

We adopt the method of Diekmann *et al.*[24] in computing the reproductive number and we define it as the number of newly infected macrophages arising out of one Mtb infected macrophage. That is we define heterogeneity using groups defined by fixed characteristics; our model can be written in the form:

$$\begin{aligned} \frac{dX}{dt} &= f(\mathbf{X}, \mathbf{Y}, \mathbf{Z}) \\ \frac{dY}{dt} &= g(\mathbf{X}, \mathbf{Y}, \mathbf{Z}) \\ \frac{dZ}{dt} &= h(\mathbf{X}, \mathbf{Y}, \mathbf{Z}) \end{aligned} \tag{11}$$

for the two categories, where $\mathbf{X} \in \Re^3$, $\mathbf{Y} \in \Re$, $\mathbf{Z} \in \Re$, and $h(\mathbf{X}, 0, 0) = 0$. Assuming that the equation $g(\mathbf{X}^*, \mathbf{Y}, \mathbf{X}) = 0$ implicitly determines a function $Y = \tilde{g}(\mathbf{X}^*, \mathbf{Y})$. We let $\mathbf{A} = \mathbf{D}_Z h(\mathbf{X}^*, \tilde{g}(\mathbf{X}^*, 0), 0)$ and further assume that $\mathbf{A}$ can be written in the form $\mathbf{A} = \mathbf{M} - \mathbf{D}$, with $\mathbf{M} \geq 0$ (i.e., $m_{ij} \geq 0$) and $\mathbf{D} > 0$, a diagonal matrix. The reproduction ratios are then evaluated from the matrix $\mathbf{MD^{-1}}$.

The cell population subgroups are divided as follows: (a) $\mathbf{X}$: are cells that are involved in immune control of Mtb infection that is; resting macrophages (M_r), health CD4+ helper T cells (T), and Mtb specific CTLs (C_b), (b) $\mathbf{Z}$: cells (macrophages) infected with the Mtb pathogen, and (c) $\mathbf{Y}$: Mtb pathogen (causative agent of TB). Therefore, we set $\mathbf{X} = (M_r, T, C_b)$, $\mathbf{Y} = Tb$, $\mathbf{Z} = M_{ib}$ and $\mathbf{X}^* = (\frac{s_m}{\mu_m}, \frac{s_t}{\mu_t}, \frac{s_b}{\mu_{cb}})$. Let $\mathbf{U}_o = (\mathbf{X}^*, 0, 0)$ denote the Mtb free equilibrium that is $f(\mathbf{X}^*, 0, 0) = g(\mathbf{X}^*, 0, 0) = h(\mathbf{X}^*, 0, 0) = 0$ and $Y = \tilde{g}(\mathbf{X}^*, \mathbf{Y})$, where

$$\begin{aligned} \tilde{g}(\mathbf{X}^*, \mathbf{Y}) = {} & \frac{k_i M_r (k_l N_c M_{ib} C_b + k_b N_b M_{ib})}{(\gamma_1 M_r + \gamma_2 C_b - r_m)} - k_b M_{ib} \\ & - k_a \frac{T M_{ib}}{M_{ib} + A} - k_l M_{ib} C_b - \mu_{mi} C_b \end{aligned} \tag{12}$$

Evaluating $\mathbf{A} = \mathbf{D}_Z h(\mathbf{X}^*, \tilde{g}(\mathbf{X}^*, 0), 0)$, we calculated that

$$\mathbf{M} = \frac{k_i \hat{M}_r (k_l N_c \hat{C}_b + k_b N_b)}{(\gamma_1 \hat{M}_r + \gamma_2 \hat{C}_b - r_m)}, \quad \mathbf{D^{-1}} = \frac{1}{k_b + k_a \frac{\hat{T}}{A} + k_l \hat{C}_b + \mu_{mi}}. \tag{13}$$

The reproduction ratio is given by computing $\rho(\mathbf{MD}^{-1})$, and it follows that

$$R_{Mtb} = \frac{k_i \hat{M}_r (k_l N_c \hat{C}_b + k_b N_b)}{(\gamma_1 \hat{M}_r + \gamma_2 \hat{C}_b - r_m)(k_b + k_a \frac{\hat{T}}{A} + k_l \hat{C}_b + \mu_{mi})}. \tag{14}$$

This expression shows that increasing the rate of infection of macrophages (k_i) and burst size (N_b) and burst rate (k_b) increases the numerical value of R_{Mtb}. An interesting result is that increasing lytic activity of CTL might tend to increase the infection progression. This might be explained by the fact that adding more bacteria (N_c) particles to the extracellular environment that favors bacterial multiplication speeds up infection progression. However, the presence of CTLs variable and parameter in the denominator indicate that CTLs are relevant in the control of Mtb infection, that is increasing lytic killing and direct killing of Mtb reduce disease progression. Also increasing the rate of extracellular bacterial multiplication increases the numerical value of R_{Mtb} hence disease progression.

Lemma 1. *The disease-free equilibrium E_{00} is locally asymptotically stable if $R_{Mtb} < 1$ and unstable if $R_{Mtb} > 1$. It is not difficult to show that E_{00} is locally asymptotically stable if $(\gamma_1 \hat{M}_r + \gamma_2 \hat{C}_b > r_m)$ using the method of linearization with aide of the Routh Hurwitz criteria.*

2.1.3. *Global stability conditions for the disease-free equilibrium*

We adopt the method of Castillo-chavez *et al.*[25] and we re-write the set of Eqs. (1)–(5) in the form:

$$\begin{aligned} \frac{dX_G}{dt} &= F(\mathbf{X_G}, \mathbf{Z_G}) \\ \frac{dZ_G}{dt} &= G(\mathbf{X_G}, \mathbf{Z_G}) \end{aligned} \tag{15}$$

With $G(X_G, 0) = 0$, where $X_G \in \Re^3$: denotes the number of uninfected cells and $Z_G \in \Re^2$: denotes the number of infected cells including latent and infectious cells. $U_{0G} = (X_G^*, 0)$ denotes the disease-free ($X_G^* = (\frac{s_m}{\mu_m}, \frac{s_t}{\mu_t}, \frac{s_b}{\mu_{cb}})$) equilibrium of the system. For the set of Eqs. (1)–(5) we set $X_G = (M_r, T, C)$: the set of resting macrophages, CD4+ helper T cells and Mtb specific CTLs and $Z_G = (Mib, Tb)$: the Mtb infected macrophages and the Mtb pathogen.

The conditions (H1) and (H2) below must be met to guarantee global asymptotic stability

(H1): for $\frac{dX_G}{dt} = F(X_G, 0)$, X_G^* is globally asymptotically stable

(H2): $G(X_G, Z_G) = A_G Z_G - \hat{G}(X_G, Z_G)$, $\hat{G}(X_G, Z_G) \geq 0$ for $(X_G, Z_G) \in \Omega$ where $A_G = D_{Z_G} G(X_G^*, 0)$ is an M-matrix (the off diagonal elements of A are non-negative) and Ω is the region where the model makes biological sense. If the above two conditions are satisfied then the following theorem holds.

Theorem 1.[25] *The fixed point* $U_{0G} = (X_G^*, 0)$ *is a globally stable equilibrium of* (15) *provided that* $R_{Mtb} < 1$ *and that assumptions* (H1) *and* (H2) *are satisfied.*

The proof of Theorem 1 and the functions $F(X_G, 0)$, A_G $(= D_{Z_G} G(X_G^*, 0))$, and $\hat{G}(X_G, Z_G) = (\hat{G}_1(X_G, Z_G), \hat{G}_2(X_G, Z_G))$ are given in Appendix A.

Therefore, if $\hat{G}(X_G, Z_G) \geq 0$ then E_{00} is globally stable. If this condition fails to hold, then E_{00} may not be globally asymptotically stable. This suggests existence of multiple equilibrium states, hence the following lemma.

Lemma 2.

(1) *If* $\hat{G}_1(X_G, Z_G) \geq 0$ *and* $\hat{G}_2(X_G, Z_G) \geq 0$, *it follows that a unique endemic equilibrium state exist.*
(2) *If either* $\hat{G}_1(X_G, Z_G) < 0$ *or* $\hat{G}_2(X_G, Z_G) < 0$, *or both are less than zero then two endemic equilibrium states might exist, OR no endemic equilibrium state exist.*

This may explain why sometimes Mtb infection gives either latent infection or active disease, or can easily be eradicated by the immune system.

2.2. *HIV model without Mtb co-infection*

We develop a model of HIV infection interaction with the immune system. We assume that when there is no co-infection, there is no pronounced infection of macrophages in the lungs with the X4 viral strain, since Mtb opportunistic infection enhance infection and multiplication of viral (X4 strain) particles in macrophages in the lungs.[18,19] In the absence of Mtb opportunistic infection, there is little or no viral replication noticed in the lungs even in patients with advanced AIDS.[18,19] The interaction of CD4+ T cells (T), infected CD4+ T cells (T^*), HIV specific CTLs (C_{hv}), and HIV virions (V) (X4 viral strain) is modeled as follows,

$$\frac{dT}{dt} = s_t + r\left(\frac{TV}{V + Aa}\right) - \beta\left(\frac{VT}{1 + a_1 C_{hv}}\right) - r_2\left(\frac{TV}{T + R}\right) - \mu_t T \tag{16}$$

$$\frac{dT^*}{dt} = \beta\left(\frac{VT}{1 + a_1 C_{hv}}\right) - h_v T^* C_{hv} - \alpha_t T^* \tag{17}$$

$$\frac{dC_{hv}}{dt} = s_v + p_v V T C_{hv} - \mu_{cv} C_{hv} \tag{18}$$

$$\frac{dV}{dt} = \left(\frac{N_v \alpha_t T^*}{1 + a_2 C_{hv}}\right) - \mu_v V. \tag{19}$$

Equation (16) models the dynamics of health CD4+ T cells. s_t is the source of CD4+ T cells from the thymus.[5] The second term represents the proliferation of CD4+ T cells due to HIV at rate r. The third and the fourth terms represent the loss of CD4+ T cells through HIV infection and apoptosis due to the bystander effect induced by HIV proteins[8] at rates β and r_2, respectively. The term $(1 + a_1 C_{hv})^{-1}$ models the inhibition of CD4+ T cells viral infection by CTLs. The last term represents the natural death of CD4+ T cells at rate μ_t.

Equation (17) models the dynamics of HIV infected CD4+ T cells (T^*). The first term is the source of infected CD4+ T cells from term 3 of Eq. (16). The second term represents the killing of infected CD4+ T cells by HIV specific CTLs (C_{hv}) at rate h_v. The last term represents both the bursting and natural death of infected CD4+ T cells at rate α_t.

Equation (18) models the dynamics of HIV specific CTLs (C_{hv}). The term s_v represents the source of CD8+ T cells from the thymus. p_v is the proliferative rate of HIV specific CTLs and μ_{cv} their natural death rate.

Equation (19) models the dynamics of the viral load. The first term represents the source of new virions from the bursting of infected CD4+ T cells. The burst size of infected CD4+ T cells is reduced by CTLs. This reduction is modelled by the term $(1 + a_2 C_{hv})^{-1}$. The last term represents the natural death of viral particles at rate μ_v.

2.2.1. *Model analysis*

The equilibrium states of this model are obtained by setting the right-hand side of Eqs. (16)–(19) to zero. The disease-free equilibrium state of Eqs. (16)–(19) are given by

$$E_{10} = (\hat{T}, \hat{T}^*, \hat{C}_{hv}, \hat{V}) = \left(\frac{s_t}{\mu_t}, 0, \frac{s_v}{\mu_{cv}}, 0\right)$$

and an HIV infected state given by

$$E_{11} = (\bar{T}, \bar{T}^*, \bar{C}_{hv}, \bar{V}),$$

where $\bar{T}$, $\bar{T}^*$, $\bar{C}_{hv}$, and $\bar{V}$ are given by expressions (20)–(23), respectively. The onset of HIV infection occurs when HIV particles enter the human body and result in the infection of CD4+ T cells that prompt the priming of HIV specific immune response. The failure of the human immune system to eliminate the HIV pathogen result in the progression of HIV infection and the depletion of CD4+ T cells.

The endemic equilibrium value of health CD4+ T cells is given by

$$\bar{T} = \frac{u_1 + \sqrt{(u_1)^2 - 4u_0 u_2}}{2u_0}, \tag{20}$$

where

$$u_0 = -\frac{r\bar{V}}{\bar{V} + A_a} + \frac{\beta\bar{V}}{1 + a_1\bar{C}_{hv}} + \mu_t,$$
$$u_1 = -\frac{rR\bar{V}}{\bar{V} + A_a} + \frac{\beta R\bar{V}}{1 + a_1\bar{C}_{hv}} + r_2\bar{V} + \mu_t R + s_t, \quad u_2 = -s_t R.$$

$u_0 > 0$, $u_1 > 0$ and $u_2 < 0$ since $\beta > r$, $(s_t, R, A_a, a_1, r, r_2, \mu_t) > 0$ and $(\bar{V} + A_a) > (1 + a_1\bar{C}_{hv})$ therefore $u_1 + \sqrt{(u_1)^2 - 4u_0u_2} \geq 0$ and expression (20) is positive. Expression (20) shows that CD4+ T cells proliferate due to the presence of the HIV pathogen, nevertheless are also depleted by the virus. The population density of infected CD4+ T cells is given by

$$\bar{T}^* = \left(\frac{\beta\bar{V}\bar{T}}{(1 + a_1\bar{C}_{hv})(h_v\bar{C}_{hv} + \alpha_t)}\right). \tag{21}$$

This expression shows that T^* will increase as infection rate β increases and will decrease as the CTL response increases. This indicate the importance of HIV specific CTLs in the control of HIV. High levels of CTLs can significantly reduce the level of infected CD4+ T cells. The equilibrium value of HIV specific CTLs is given by

$$\bar{C}_{hv} = \left(\frac{s_v}{\mu_{cv} - p_v\bar{T}\bar{V}}\right). \tag{22}$$

This expression shows that there is an inverse relationship between V and T, that is, for $\bar{C}_{hv}$ to be positive, $\mu_{cv} > p_v\bar{T}\bar{V}$, $(\bar{V} < \left(\frac{\mu_{cv}}{p_v\bar{T}}\right), \Rightarrow \bar{V} \propto \frac{1}{\bar{T}})$. This implies that increased viral growth results in high depletion of CD4+ T cells. The viral steady state is given by

$$\bar{V} = \left(\frac{N_v\alpha_t\bar{T}^*}{\mu_v(1 + a_2\bar{C}_{hv})}\right). \tag{23}$$

This shows that viral load increases as the burst sizes and burst rate of T^* increase and is reduced by HIV specific CTL response. Increasing infection of CD4+ T cells result in increased viral production. In light of expression (21), HIV specific CTLs are really necessary in the control of infection, since high volumes of CTLs can suppress viral production.

2.2.2. *Reproduction ratio*

We calculate the reproduction number using the approach we used in Sec. 2.1.2 and we define the reproduction number as the number of newly infected CD4+ T cells arising from one HIV infected CD4+ T cell. Using the same method we define heterogeneity using groups defined by fixed characteristics; our model can be written in the form (11), where $\mathbf{X_A} \in \Re^2$, $\mathbf{Y_A} \in \Re$, $\mathbf{Z_A} \in \Re$, and $h_A(\mathbf{X_A}, 0, 0) = 0$. We follow the same procedure as we did in computing the reproduction ratio for Mtb infection only. Assuming that the equation $g_A(\mathbf{X_A^*}, \mathbf{Y_A}, \mathbf{X_A}) = 0$ implicitly determines a function $Y_A = \tilde{g}_A(\mathbf{X_A^*}, \mathbf{Y_A})$. We let $\mathbf{A_A} = \mathbf{D}_{Z_A}h_A(\mathbf{X_A^*}, \tilde{g}_A(\mathbf{X_A^*}, 0), 0)$ and further assume that $\mathbf{A_A}$ can be written in the form $\mathbf{A_A} = \mathbf{M_A} - \mathbf{D_A}$, with $\mathbf{M_A} \geq 0$ (i.e. $m_{ij} \geq 0$) and $\mathbf{D_A} > 0$, a diagonal matrix. The reproduction ratios are then evaluated from the matrix $\mathbf{M_A}\mathbf{D_A^{-1}}$.

The cell population subgroups are divided as follows, (a) $\mathbf{X_A}$: are cells that are uninfected by the virus (T, C_{hv}). (b) $\mathbf{Z_A}$: cells that are virus infected (infected CD4+ T cells), and (c) $\mathbf{Y_A}$: the HIV pathogen. Therefore, we set $\mathbf{X_A} = (T, C_{hv})$, $\mathbf{Y_A} = V$, $\mathbf{Z_A} = T^*$, and $\mathbf{X_A^*} = (\frac{s_t}{\mu_t}, \frac{s_v}{\mu_{cv}})$. Let $\mathbf{U}_{0A} = (\mathbf{X_A}^*, 0, 0)$ denote the virus free equilibrium, that is $f_A(\mathbf{X_A}^*, 0, 0) = g_A(\mathbf{X_A}^*, 0, 0) = h_A(\mathbf{X_A}^*, 0, 0) = 0$ and $Y_A = \tilde{g}_A(\mathbf{X_A}^*, \mathbf{Z_A})$, where

$$\tilde{g}_A(\mathbf{X_A}^*, \mathbf{Z_A}) = \frac{N_v \alpha_t T^*}{\mu_v(1 + a_2 C_{hv})}.$$

We computed $A_A = D_{Z_A}(X_A^*, \tilde{g}_A(X_A^*, 0), 0)$ and we evaluated that $M_A = \frac{\beta \hat{T} N_v \alpha_t}{\mu_v(1+a_1\hat{C}_{hv})(1+a_2\hat{C}_{hv})}$ and $D_A^{-1} = \frac{1}{(h_v\hat{C}_{hv}+\alpha_t)}$. The preproduction ratio is given by the next generation spectral radius $\rho(M_A D_A^{-1})$ as follows,

$$R_{\text{HIV}} = \frac{N_v \alpha_t \beta \hat{T}}{\mu_v(h_v \hat{C}_{hv} + \alpha_t)(1 + a_1 \hat{C}_{hv})(1 + a_2 \hat{C}_{hv})} \tag{24}$$

Reducing R_{HIV} to a value less than one controls infection. The expression for R_{HIV} (24) suggest that enhancing the activity of CTLs reduce disease progression while increasing the infection rate (β), burst size (N_v) and burst rate (α_t) of CD4+ T cells accelerates infection progression.

Lemma 3. *The disease-free equilibrium E_{10} is locally asymptotically stable if $R_{\text{HIV}} < 1$ and unstable if $R_{\text{HIV}} > 1$. This can be easily shown via the method of linearization with aid of the Routh Hurwitz criteria.*

2.2.3. *Global stability conditions for the disease free equilibrium*

Applying the same method we used in Sec. 2, we can also determine the stability of the model equilibrium states.

The set of Eqs. (16)–(19) can be written in the form (15), with $G_B(X_B, 0) = 0$, $X_B = \Re^2$, and $Z_B = \Re^2$. We set $X_B = (T, C_{hv})$: the set of CD4+ helper T cells and HIV specific CTLs and $Z_B = (T^*, V)$: the HIV infected CD4+ T cells and the HIV pathogen. The computations of $F_B(X_B, 0)$, A_B $(= D_{Z_B} G_B(X_B^*, 0))$ and $\hat{G}_B(X_B, Z_B)$ are given in Appendix B, where $X_B^* = (\frac{s_t}{\mu_t}, \frac{s_v}{\mu_{cv}})$ and the following theorem whose proof is given in Appendix B follows,

Theorem 2.[25] *The fixed point $U_{0B} = (X_B^*, 0)$ is a globally stable equilibrium of* (15) *provided that $R_{\text{HIV}} < 1$ and that assumptions* (H1) *and* (H2) *of Theorem* 1 *are satisfied.*

E_{10} is globally asymptotically stable since the analysis in Appendix B shows that $\hat{G}_B(X_B, Z_B) \geq 0$.

3. Co-infection Model Equations

This model considers the co-infection of HIV/Mtb infections. We model the infection of macrophages by both the Mtb and HIV pathogens. A single HIV strain (X4 strain) is used in-order to simplify the model; macrophages are predominantly infected by the R5 strain that tends to mutate to the X4 strain as infection progress.

But their infection with the X4 strain is significantly noticed in the lungs during TB and HIV co-infection. Howbeit, in the absence of co-infection little or no viral multiplication is not noticed in the lungs. CD4+ T helper cells are predominantly infected by the X4 strain which is mainly associated with fast progression of HIV infection to the AIDS stage.[18,19]

$$\frac{dM_r}{dt} = s_m + p_m(r_o V + T_b)M_r - k_v\left(\frac{VM_r}{1 + a_o C_{hv}}\right) - k_i T_b M_r - \mu_m M_r \tag{25}$$

$$\frac{dM_{ib}}{dt} = k_i T_b M_r - k_b M_{ib} - k_a\left(\frac{M_{ib}}{M_{ib} + A}\right)T - k_l M_{ib} C_b - \mu_{mi} M_{ib} \tag{26}$$

$$\frac{dM_{iv}}{dt} = k_v\left(\frac{VM_r}{1 + a_o C_{hv}}\right) - h_m M_{iv} C_{hv} - m_b M_{iv} \tag{27}$$

$$\frac{dT_b}{dt} = k_b N_b M_{ib} - \gamma_1 T_b M_r - \gamma_2 T_b C_b + r_m T_b + k_l N_c M_{ib} C_b \tag{28}$$

$$\begin{aligned}\frac{dT}{dt} &= s_t + r\left(\frac{TV}{V + Aa}\right) + r_1\left(\frac{TT_b}{T_b + B}\right)\\ &\quad - \beta\left(\frac{VT}{1 + a_1 C_{hv}}\right) - r_2\left(\frac{TV}{T + R}\right) - \mu_t T\end{aligned} \tag{29}$$

$$\frac{dT^*}{dt} = \beta\left(\frac{VT}{1 + a_1 C_{hv}}\right) - h_v T^* C_{hv} - \alpha_t T^* \tag{30}$$

$$\frac{dC_{hv}}{dt} = s_v + p_v VT C_{hv} - \mu_{cv} C_{hv} \tag{31}$$

$$\frac{dC_b}{dt} = s_b + p_b\left(\frac{M_{ib}}{M_{ib} + R_b}\right)TC_b - \mu_{cb} C_b \tag{32}$$

$$\frac{dV}{dt} = \left(\frac{N_v \alpha_t T^*}{1 + a_2 C_{hv}}\right) + \left(\frac{N_m m_b M_{iv}}{1 + a_3 C_{hv}}\right) - \mu_v V. \tag{33}$$

Equation (25) models the dynamics of resting macrophages (M_r). Here s_m is the source of macrophages from bone marrow progenitor monocyte differentiation.[22] New macrophages are recruited in response to HIV and Mtb infections at rates $p_m r_o$ and p_m, where r_o differentiates the rate of resting macrophages recruitment in response to Mtb and HIV infections. The third term of Eq. (25) represents the infection of resting macrophages by the virus[7,18] at rate k_v. The term $(1+a_o C_{hv})^{-1}$ models inhibition of resting macrophages infection by the virus. a_o is the inhibition factor. k_i represents the rate at which Mtb (T_b) infects resting macrophages. The last term represents the natural death of resting macrophages.

Equation (27) models the dynamics of HIV infected macrophages (M_{iv}) that are infected by HIV.[7,18,19] The first term represents the source of HIV infected macrophages from term 4 of Eq. (25). HIV infected macrophages are killed by HIV specific CTLs (C_{hv}) at rate h_m and burst at rate m_b as the viral load increase. In this model, we incorporated the natural death of HIV infected macrophages in m_b.

Equation (29) models the dynamics of health CD4+ T cells. s_t is the source of CD4+ T cells from the thymus.[5] The second and third terms represent the proliferation of CD4+ T cells due to HIV and Mtb infections at proliferative rates r and r_1, respectively. The fourth and the fifth terms represent the depletion of CD4+ T cells through HIV infection and apoptosis due to the bystander effect induced by HIV proteins[8] at rates β and r_2, respectively. The term $(1 + a_1 C_{hv})^{-1}$ models the inhibition of CD4+ T cells infection by viral particles. The last term represents the natural death of CD4+ T cells at rate μ_t.

Equation (30) models the dynamics of HIV infected CD4+ T cells (T^*). The first term is the source of infected CD4+ T cells from term 4 of Eq. (29). The second term represents the killing of infected CD4+ T cells by HIV specific CTLs (C_{hv}) at rate h_v. The last term represents the bursting of infected CD4+ T cells at rate α_t.

Equation (33) models the dynamics of the viral load. The first and the second terms represent the source of new virions from the bursting of infected CD4+ T cells and HIV infected macrophages, respectively. The burst size of infected CD4+ T cells and HIV infected macrophages are reduced by CTLs. This reduction is modeled by the terms $(1+a_2C_{hv})^{-1}$ and $(1+a_3C_{hv})^{-1}$, respectively. The last term represents the natural death of viral particles at rate μ_v.

Equations (26), (28), (31), and (32) models the dynamics of Mtb infected macrophages, Mtb pathogen, HIV specific CTLs and Mtb specifc CTLs during TB and HIV co-infection, respectively. These equations are basically explained in the same way as Eqs. (2), (3), (18), and (5) of the Mtb and HIV separate models, respectively.

3.1. *Analysis of the model*

We analyse system of Eqs. (25)–(33) by determining its steady states. The dynamical system has an uninfected state given by

$$E_{20} = (\hat{M}_r, \hat{M}_{ib}, \hat{M}_{iv}, \hat{T}_b, \hat{T}, \hat{T}^*, \hat{C}_{hv}, \hat{C}_b, \hat{V}) = \left(\frac{s_m}{\mu_m}, 0, 0, 0, \frac{s_t}{\mu_t}, 0, \frac{s_v}{\mu_{cv}}, \frac{s_b}{\mu_{cb}}, 0\right).$$

If infection persists after initial inoculum the system converges to an infected steady state given by

$$E_{21} = (\bar{M}_r, \bar{M}_{ib}, \bar{M}_{iv}, \bar{T}_b, \bar{T}, \bar{T}^*, \bar{C}_{hv}, \bar{C}_b, \bar{V}),$$

where $\bar{M}_r$, $\bar{M}_{ib}$, $\bar{M}_{iv}$, $\bar{T}_b$, $\bar{T}$, $\bar{T}^*$, $\bar{C}_{hv}$, $\bar{C}_b$, and $\bar{V}$ are given by expressions (34)–(42), respectively. This co-infection equilibrium state represents the scenario where both the Mtb and HIV infections are present. However, it is possible to represent two other separate scenarios when one of the infections is absent, that is, HIV infection without Mtb infection or vise-versa. Mtb/HIV co-infection accelerates Mtb infection and activates TB latency to active disease.

The infected state of resting macrophages is given by

$$\bar{M}_r = \frac{s_m}{\left(\frac{k_v \bar{V}}{1+a_o \bar{C}_{hv}}\right) + k_i \bar{T}_b + \mu_m - p_m(r_o \bar{V} + \bar{T}_b)}. \tag{34}$$

Expression (34) shows that the equilibrium value of M_r depends on both the HIV and Mtb pathogens. Comparing expression (6) with (34) recovering macrophages count in (34) requires higher recruitment rates during co-infection. Infection of macrophages by HIV in the lungs reduces resting macrophages (M_r) count. This favor fast TB progression. The population of Mtb infected macrophages is given by

$$\bar{M}_{ib} = \frac{-w_0 + \sqrt{w_0^2 - 4w_1 w_2}}{2w_1}, \tag{35}$$

where

$$w_0 = k_i \bar{T}_b \bar{M}_r - k_b A - k_a - k_l \bar{C}_b - \mu_{mi} A,$$
$$w_1 = k_b + k_l \bar{C}_b + \mu_{mi}, \quad w_2 = k_i \bar{T}_b \bar{M}_r A.$$

This shows that the equilibrium value of Mtb infected macrophages. This expression is similar to expression (7) when there is Mtb infection alone, only that these dynamics are during co-infection. The steady state value of macrophages that are infected by the virus is given by

$$\bar{M}_{iv} = \left(\frac{k_v \bar{V} \bar{M}_r}{(h_m \bar{C}_{hv} + m_b)(1 + a_o \bar{C}_{hv})}\right). \tag{36}$$

This expression shows that the population of M_{iv} will increase if the rate of M_r infection by HIV-1 increases and will decrease as the effectiveness of CTLs (lytic killing and infection inhibition) increase. However, macrophages are insensitive to cytopathic effects of HIV and depletion of T cells by the virus makes proliferation of CTLs poor. These mechanisms selectively favor production of HIV virions in macrophages for protracted periods of time. The bacterial density at equilibrium is given as follows

$$\bar{T}_b = \left(\frac{k_b N_b \bar{M}_{ib} + k_l N_c M_{ib} C_b}{\gamma_1 \bar{M}_r + \gamma_2 \bar{C}_b - r_m}\right). \tag{37}$$

This expression is similar to (8) in the absence of co-infection. Conditions in co-infection favor multiplication of Mtb bacterial since CTL proliferation (dependent on CD4+ T cells) is poor and resting macrophages that are central in the control of Mtb infection are also infected by the virus. Therefore this expression (37) will attain a higher numerical value than the value of expression (8). The equilibrium value of health CD4+ T cells is given by

$$\bar{T} = \left(\frac{-(s_t + r f_o - r_2 \bar{V}) + \sqrt{(s_t + r f_o - r_2 \bar{V})^2 - 4 f_o s_t R}}{2 f_o}\right), \tag{38}$$

where

$$f_o = \left(\frac{r\bar{V}}{\bar{V} + A_a} + \frac{r_1 \bar{T}_b}{\bar{T}_b + B} - \frac{\beta \bar{V}}{1 + a_1 \bar{C}_{hv}} - \mu_t\right).$$

This shows that CD4+ T cells increase in population due to the presence of TB and HIV pathogens, and are also depleted by the virus. The population density of infected CD4+ T cells is given by

$$\bar{T}^* = \left(\frac{\beta\bar{V}\bar{T}}{(1 + a_1\bar{C}_{hv})(h_v\bar{C}_{hv} + \alpha_t)}\right). \tag{39}$$

This expression is similar to (21). It increases as infection rate β increases and decreases as the CTL response increases. In this expression $\bar{V}$ is greater than in (21), therefore during co-infection there are more infected CD4+ T cells than when there is no co-infection. The equilibrium value of HIV specific CTLs is given by

$$\bar{C}_{hv} = \left(\frac{s_v}{\mu_{cv} - p_v\bar{T}\bar{V}}\right). \tag{40}$$

This expression shows that there is an inverse relationship between $\bar{V}$ and $\bar{T}$, as we have shown in the case of HIV infection only. Comparing this expression with (22), here $\bar{C}_{hv}$ will have a smaller value since $\bar{V}$ (given by (42)) assume a bigger value due to the contribution of viral particles from infected macrophages. The steady-state value of Mtb specific CTLs is given by

$$\bar{C}_b = \left(\frac{s_b}{\mu_{cb} - p_b\left(\frac{\bar{M}_{ib}}{\bar{M}_{ib}+R_b}\right)\bar{T}}\right). \tag{41}$$

This expression is similar to expression (10), however here Mtb specific CTLs levels are lower because their proliferation depends on CD4+ T cells that are heavily depleted by HIV during co-infection. The viral steady state is given by

$$\bar{V} = \left(\frac{N_v\alpha_t\bar{T}^*}{\mu_v(1 + a_2\bar{C}_{hv})} + \frac{N_m m_b\bar{M}_{iv}}{\mu_v(1 + a_3\bar{C}_{hv})}\right). \tag{42}$$

This shows that viral load increases as the burst sizes of T^* and M_{iv} increase and is reduced by HIV specific CTL response. The addition of viral particles from infected macrophages amounts to more viral particles in the lungs. This would result in pronounced infection of macrophages by the virus leading to fast TB progression. This also suggest that there will be continual production of virus even if there is total depletion of CD4+ T cells.

It is imperative to notice that in this model the proliferation of both Mtb specific CTLs and HIV specific CTLs depend on the same pool of CD4+ T cells. However, the dynamics of HIV and Mtb specific CTLs are different, since CTLs are quite specific to different invading pathogens. The cytokine and chemokine profiles required for the priming of each pathogen's specific immune response is different. It might happen that the cytokine milieu generated in response to Mtb infection and necessary for the priming of Mtb specific immune response might support the replication of HIV particles. One of such is TNF-α and has been found to be associated with high HIV replication.[2,3] The levels of CTL expression at the site of co-infection should be a reflection/indicator of the balance/imbalance/expression of cytokine profiles exhibited by the Mtb and HIV pathogens. Therefore, we hypothesize that the ratio of Mtb specific CTLs expression to HIV specific CTLs should be proportional to the ratio of Mtb bacterial burden to viral load in the lungs or the dynamics of HIV viral particles to Mtb bacteria at the site of infection. The idea of competitive exclusion here is out of question because the Mtb pathogen and HIV pathogen are two different species and their mechanisms of reproduction, the type of cells they infect are different and they cause different diseases. Further more each pathogen can cause a disease in the absence of the other. These pathogens can coexist and they are known to support the multiplication of each other. This hypothesis follows from analyzing Eqs. (31) and (32) in light of expressions (40) and (41). Rearranging these expressions we get

$$\left(\frac{1}{\bar{C}_b}\right) = \left(\frac{\mu_{cb}}{s_b}\right) - \left(\frac{p_b}{s_b}\right)\bar{T}\bar{T}_B \tag{43}$$

$$\left(\frac{1}{\bar{C}_{hv}}\right) = \left(\frac{\mu_{cv}}{s_v}\right) - \left(\frac{p_v}{s_v}\right)\bar{T}\bar{V} \tag{44}$$

where $\bar{T}_B = \frac{\bar{M}_{ib}}{\bar{M}_{ib}+R_b}$. $\bar{T}_B$ is the density of infected macrophages that indicate/reflect the cytokine milieu induced by Mtb infection that is responsible for Mtb specific CTL proliferation; $\bar{T}_B$ is also an indicator of Mtb infection severity. Making $\bar{T}$ subject to Eq. (44) and substituting the result for $\bar{T}$ in (43) yields the following relationship

$$\left(\frac{1}{\bar{C}_b}\right) = \left(\frac{\mu_{cb}}{s_b}\right) + \left(\frac{p_b}{s_b}\right)\left(\frac{p_v}{s_v}\right)\left(\frac{\bar{T}_B}{\bar{V}}\right)\left(\frac{1}{\bar{C}_{hv}}\right) - \left(\frac{p_b}{s_b}\right)\left(\frac{s_v}{p_v}\right)\left(\frac{\mu_{cv}}{s_v}\right)\left(\frac{\bar{T}_B}{\bar{V}}\right) \tag{45}$$

This relationship shows that the expression of Mtb specific CTLs is related partially (i) to the expression of HIV CTLs and (ii) ratio of the density of TB and HIV. The following points can be noted from this expression

(1) Any factor that reduces the RHS of (45) increases the expression of Mtb specific CTLs.
(2) Increasing HIV specific CTLs ($\bar{C}_{hv}$) increases the Mtb specific CTLs. High HIV CTL levels are known to be associated with less viral load and less depletion of CD4+ T cells, hence more CD4+ T cells to support the development of Mtb immune response.
(3) Increasing the natural turn over $\left(\frac{s_b}{\mu_{cb}}\right)$ of Mtb CTLs increase their expression at the site of infection.
(4) It is not straight forward to interpret the terms $\left(\frac{p_b}{s_b}\right)\left(\frac{p_v}{s_v}\right)\left(\frac{\bar{T}_B}{\bar{V}}\right)\left(\frac{1}{\bar{C}_{hv}}\right)$ and $\left(\frac{p_b}{s_b}\right)\left(\frac{s_v}{p_v}\right)\left(\frac{\mu_{cv}}{s_v}\right)\left(\frac{\bar{T}_B}{\bar{V}}\right)$, but they seem to suggest that the level of Mtb CTLs are somehow related to the ratio of Mtb bacteria to HIV viral load.

Considering the fourth point further, we notice that $\left(\frac{1}{\bar{C}_b}\right)$ varies partially with $\left(\frac{\bar{T}_B}{\bar{V}}\right)\left(\frac{1}{\bar{C}_{hv}}\right)$ and $\left(\frac{\bar{T}_B}{\bar{V}}\right)$, that is, (i) $\left(\frac{1}{\bar{C}_b}\right) \propto \left(\frac{\bar{T}_B}{\bar{V}}\right)$ and (ii) $\left(\frac{1}{\bar{C}_b}\right) \propto \left(\frac{\bar{T}_B}{\bar{V}}\right)\left(\frac{1}{\bar{C}_{hv}}\right)$. (i) suggest that $\bar{C}_b$ varies inversely with $\bar{T}_B$, but there is more insight if we say $\bar{C}_b$ varies directly to $(\frac{\bar{V}}{\bar{T}_B})$. This implies that as the virus increases TB increases and Mtb CTLs decrease since increase in the virus impairs the immune system. On the other hand, (ii) suggest that as HIV specific CTLs increase Mtb CTL expression also increases. Increase in HIV CTLs is coupled with reduced viral load and reduced CD4+ T cell depletion hence boasting of the T cells count that is necessary for the development of Mtb specific immunity. Moreover, this relationship is dependent on the ratio $(\frac{\bar{V}}{\bar{T}_B})$. This seems to suggest that increase in the virus result in the increase of Mtb CTLs. But there is no direct way the virus is related to Mtb CTLs and will not make sense to say that increase in the virus will increase Mtb CTLs, therefore it seems more plausible to return to our explanation of (i). Nevertheless expression (ii) can be rearranged to $(\frac{\bar{C}_b}{\bar{C}_{hv}}) \propto (\frac{\bar{V}}{\bar{T}_B})$, which makes sense in that $\bar{C}_{hv} \propto \frac{1}{\bar{V}}$ and $\bar{C}_b \propto \frac{1}{\bar{T}_B}$ during co-infection. This shows that the ratio of Mtb CTLs to HIV CTLs is proportional to the ratio of viral load to Mtb bacterial burden at the site of infection. Since Mtb CTLs are inversely proportional to TB and HIV CTLs are also inversely proportional to the virus. From this analysis we establish our hypothesis that Mtb and HIV CTL expression during co-infection is inversely proportional to the Mtb and HIV pathogen densities at the site of infection, that is, $\left(M_{tb}HIV\ CTLs \propto \frac{1}{(M_{tb}HIV_{\text{Densities}})}\right)$. This means that as Mtb and HIV CTLs increase the Mtb and HIV pathogens decrease and vice-versa. Increase in Mtb CTLs will drive the Mtb pathogen low, hence reduction in the Mtb HIV synergy that support rapid HIV replication.

3.1.1. *Reproduction ratio*

The reproduction ratio in this case, where there is HIV and Mtb co-infection is defined by the spectral radius of the next generation approach. This gives rise to two reproduction ratios (for Mtb and HIV) since this is a co-infection of two pathogens that are capable of causing infection in the absence of the other.

We use the next generation approach to compute R_0.[24] Let $\mathcal{F}_i$ be the rate of new infections into compartment i, $\mathcal{V}_i^+(x)$ be the rate of transfer of susceptible cells into i by all other means and $\mathcal{V}_i^-(x)$ be the transfer out of i ($\mathcal{V}_i = \mathcal{V}_i^- - \mathcal{V}_i+$). For the system of Eqs. (25)–(33) $\mathcal{F}$ and $\mathcal{V}$ are given as follows:

$$\mathcal{F} = \begin{pmatrix} k_i T_b M_r \\ \left(\frac{k_v V M_r}{1+a_o C_{hv}}\right) \\ \left(\frac{\beta V T}{1+a_1 C_{hv}}\right) \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix},$$

$$\mathcal{V} = \begin{pmatrix} k_b Mi_b + k_a\left(\frac{Mi_b}{Mi_b+A}\right)T + k_l Mi_b C_b + \mu_{mi} Mi_b \\ h_m Mi_v C_{hv} + m_b Mi_v \\ h_v T^* C_{hv} + \alpha_t T^* \\ \gamma_1 T_b M_r + \gamma_2 T_b C_b - k_b N_b Mi_b - r_m T_b - k_l N_c M_{ib} C_b \\ \mu_v V - \left(\frac{N_v \alpha_t T^*}{1+a_2 C_{hv}}\right) - \left(\frac{N_m m_b Mi_v}{1+a_3 C_{hv}}\right) \\ k_v\left(\frac{VM_r}{1+a_o C_{hv}}\right) + k_i T_b M_r + \mu_m M_r - s_m - p_m(r_o V + T_b)M_r \\ \beta\left(\frac{VT}{1+a_1 C_{hv}}\right) + r_2\left(\frac{TV}{T+R}\right) + \mu_t T - s_t - r\left(\frac{TV}{V+A_a}\right) - r_1\left(\frac{TT_b}{T_b+B}\right) \end{pmatrix}.$$

The derivatives of $\mathcal{F}$, $(\mathcal{DF}(E_{20}))$ and $\mathcal{V}$, $(\mathcal{DV}(E_{20}))$ are evaluated and partitioned to give F_a and V_a that are 5×5 matrices. To find the reproduction ratio, we evaluate the eigenvalues of the matrix given by $F_a V_a^{-1}$. Therefore, $R_0 = \rho(F_a V_a^{-1})$, where $\rho(F_a V_a^{-1})$ denotes the spectral radius of matrix $(F_a V_a^{-1})$. When evaluated this gives the following reproduction ratios, R_{Tb_c} and R_{HIV_c}. The values of R_{Tb_c} and R_{HIV_c} give the reproduction ratios of TB and HIV infections during co-infection, respectively and are given as follows:

$$R_{Tb_c} = \left(\frac{k_i \hat{M}_r (k_b N_b + k_l N_c \hat{C}_b)}{(\gamma_1 \hat{M}_r + \gamma_2 \hat{C}_b - r_m)(k_b + k_a\left(\frac{\hat{T}}{A}\right) + k_l \hat{C}_b + \mu_{mi})}\right), \tag{46}$$

$$R_{\mathrm{HIV}_c} = \frac{1}{\mu_v}\left(\frac{N_m m_b k_v \hat{M}_r}{(1+a_3\hat{C}_{hv})(1+a_o\hat{C}_{hv})(h_m\hat{C}_{hv}+m_b)} + \frac{N_v \alpha_t \beta \hat{T}}{(1+a_1\hat{C}_{hv})(1+a_2\hat{C}_{hv})(h_v\hat{C}_{hv}+\alpha_t)}\right). \tag{47}$$

Lemma 4.

(1) *If* $R_{Tb_c}, R_{\mathrm{HIV}_c} < 1$, *then the disease-free equilibrium* E_{20} *is locally asymptotically stable.*
(2) *If* $R_{Tb_c} > 1$ *and* $R_{\mathrm{HIV}_c} < 1$, *then the disease-free equilibrium will be unstable and will result in an Mtb infected endemic state.*
(3) *If* $R_{Tb_c} < 1$ *and* $R_{\mathrm{HIV}_c} > 1$, E_{20} *will not be stable again but in this case will give an HIV infected endemic state.*
(4) *If both* $R_{Tb_c}, R_{\mathrm{HIV}_c} > 1$, *the disease-free state will be unstable and it will give an Mtb/HIV co-infected state.*

It can be shown via linearization that the disease-free equilibrium is locally stable if $(\gamma_1 \hat{M}_r + \gamma_2 \hat{C}_b > r_m)$ with aide of Routh Hurwitz criteria.

If $R_{Tb_c}, R_{\text{HIV}_c} < 1$ then the Mtb and HIV infections are abortive and if $R_{Tb_c}, R_{\text{HIV}_c} > 1$ then Mtb infection would progress to give either latent or active TB, while HIV infection would slowly/rapidly proceed until AIDS is achieved. These have profound impact on each other and they are linked to each other through resting macrophages and health CD4+ T cells. Expressions (40) and (41) show an inverse relationship between V (viral particles), M_{iv} (HIV infected macrophages), and M_{ib} (Mtb infected macrophages) levels to health CD4+ T cell levels. HIV results in depletion of CD4+ T cells and macrophages that are instrumental in the control of Mtb infection thus worsening the picture of Mtb infection. On the other hand Mtb infection results in increased rate of HIV replication and in the depletion of macrophages. This fuels the HIV infection. Since these two reproduction ratios are linked through CD4+ T cells and macrophages, this shows that the TB/HIV infection is a deadly combination since each disease progression fuels the progression of the other. R_{Tb_c} is similar to R_{Mtb}, the difference is in the value of $\hat{T}$ since in co-infection CD4+ T cells are depleted by the virus. Depletion of CD4+ T cells by the virus during co-infection makes the priming and proliferation of Mtb specific immune system impaired. R_{HIV_c} is different from R_{HIV} because of the addition of viral particles from infected macrophages. These reproduction numbers suggest that the progression of HIV and TB in TB/HIV co-infection goes a step ahead just the depletion of CD4+ T cells. Infection of macrophages can be a source of continual viral production even after total depletion of CD4+ T cells.

3.1.2. *Global stability during co-infection at the disease free equilibrium state*

Again using the approach in Sec. 2, we can establish the global stability of the co-infection model. We can re-write the set of Eqs. (25)–(33) in the form (15), with $G_C(X_C, Z_C) = 0$, $X_C = \Re^4$, and $Z_C = \Re^5$. We set $X_C = (M_r, T, C_{hv}, C_b)$, $(X_C^* = (\frac{s_m}{\mu_m}, \frac{s_t}{\mu_t}, \frac{s_v}{\mu_{cv}}, \frac{s_b}{\mu_{cb}}))$: the set of resting macrophages, CD4+ helper T cells, HIV specific CTLs, and Mtb specific CTLs and $Z_C = (M_{ib}, M_{iv}, T^*, T_b, V)$: the Mtb infected macrophages, HIV infected macrophages, HIV infected CD4+ T cells, Mtb pathogen, and the HIV pathogen. $F_C(X_C, 0)$, A_C $(= D_{Z_C} G_C(X_C^*, 0))$, and $\hat{G}_C(X_C, Z_C)$ calculations are given in Appendix C.

Theorem 3.[25] *The fixed point* $U_{0C} = (X_C^*, 0)$ *is a globally stable equilibrium of* (15) *provided that* $R_{Tb_c} < 1$ *and* $R_{\text{HIV}_c} < 1$, *and that assumptions* (H1) *and* (H2) *of Theorem* 1 *are satisfied.*

Proof is given in Appendix C.

Following the analysis in Appendix C, E_{20} is globally asymptotically stable if $\hat{G}_C(X_C, Z_C) \geq 0$. If this condition is not satisfied, then E_{20} may not be globally asymptotically stable. Therefore, the same lemma (*lemma* 1) and conclusions from the model of Mtb infection only follows.

3.1.3. *Comparison of reproduction numbers of Mtb and HIV infections with and without co-infection*

The HIV reproduction ratio during co-infection can be re-written as follows:

$$R_{\mathrm{HIV}_c} = R_{\mathrm{HIV}} + R_{V_M}, \tag{48}$$

where

$$R_{V_M} = \left(\frac{N_m m_b k_v \hat{M}_r}{\mu_v (h_m \hat{C}_{hv} + \alpha_t)(1 + a_0 \hat{C}_{hv})(1 + a_3 \hat{C}_{hv})} \right).$$

The HIV reproduction number R_{HIV_c} during co-infection is greater than the HIV reproduction number R_{HIV} when there is no co-infection (Table 1). R_{HIV_c} is the sum of R_{HIV} and R_{V_M}. $R_{\mathrm{HIV}_c} = R_{V_T}$ when there is no Mtb co-infection; co-infection makes resting macrophages more susceptible to HIV infection.[18,19] R_{V_M} is the contribution to the HIV reproduction ratio due to the infection of macrophages by HIV. It was reported that there is noticeable rapid replication of viral particles in the lungs during co-infection which is not noticed when there is no co-infection.[2–4] In the absence of an opportunistic infections, there is little or no viral replication in the lungs even in patients with advanced AIDS[18] and macrophages are the major cell type in which HIV-1 replication occur.[18,19,26] This may underlie the increased mortality observed in patients co-infected with HIV-1 and TB. R_{V_M} increases as k_v (rate at which the virus infects resting macrophages) and it is also increased by increasing burst size (N_m), burst rate (m_b). Increased CTLs mechanisms that inhibit macrophage infection by HIV, virion production and lytic killing reduces the value of R_{V_M}. This is a keynote since the activity of CTLs on the infection of macrophages by HIV virions is poor. Macrophages offer a protected environment for multiplication of viral particles.[7,18,19,27,28] This suggests that another way to reduce HIV

Table 1. Reproduction numbers.

R_o without co-infection	R_0 with co-infection
$R_{Mtb} = \left(\frac{k_i \hat{M}_r (k_l N_c \hat{C}_b + k_b N_b)}{(\gamma_1 \hat{M}_r + \gamma_2 \hat{C}_b - r_m) \times \left(k_b + k_a \frac{\hat{T}_1}{A} + k_l \hat{C}_b + \mu_{mi}\right)} \right)$	$R_{Tb_c} = \left(\frac{k_i \hat{M}_r (k_l N_c \hat{C}_b + k_b N_b)}{(\gamma_1 \hat{M}_r + \gamma_2 \hat{C}_b - r_m) \times \left(k_b + k_a \frac{\hat{T}}{A} + k_l \hat{C}_b + \mu_{mi}\right)} \right)$
$R_{\mathrm{HIV}} = \left(\frac{N_v \alpha_t \beta \hat{T}_2}{\mu_v (h_v \hat{C}_{hv} + \alpha_t)(1 + a_1 \hat{C}_{hv}) \times (1 + a_2 \hat{C}_{hv})} \right)$	$R_{\mathrm{HIV}_c} = \left(\frac{N_m m_b k_v \hat{M}_r}{\mu_v (h_m \hat{C}_{hv} + \alpha_t)(1 + a_0 \hat{C}_{hv}) \times (1 + a_3 \hat{C}_{hv})} \right) + \left(\frac{N_v \alpha_t \beta \hat{T}_2}{\mu_v (h_v \hat{C}_{hv} + \alpha_t)(1 + a_1 \hat{C}_{hv}) \times (1 + a_2 \hat{C}_{hv})} \right)$

infection progression to AIDS stage is to reduce infection of macrophages by the virus, that is forcing R_{V_M} to approach zero. A lot of Anti-retrovirals are on the market that reduce infection of CD4+ T cells and production of viral particles through the use of protease inhibitors (PIs) and reverse transcriptase inhibitors (RTIs). The contribution to the viral load from HIV infected macrophages is also of importance since there are many sanctuaries where HIV particles are hiden inside macrophages which are poorly penetrated by HIV drugs.[29,30] This might be the reason behind failure of eradication of HIV through anti-retrovirals and what seems to be viral rebound after reduction of viral load through use of anti-retroviral therapy.[27] HIV infected macrophages might serve as the reservoir of viral particles.

It is not easy to show the effect of the HIV co-infection on the Mtb reproduction number analytically. But our numerical simulations show the impact of co-infection on the progression of Mtb infection. During co-infection latent Mtb is easily activated to active TB. This is because the T cells that keep Mtb infection in check are rendered dysfunctional.[1,31] It is known that the hallmark of HIV infection is the depletion of CD4+ T cells. It has been shown in Refs. 12, 15 and 22 that T cells are key cells in mounting a robust Mtb specific immune response. Therefore, depletion of T cells result in poor Mtb immune response, hence fast progression of the infection. Also HIV infect macrophages during co-infection which are also keys cell in controlling Mtb infection.

If we assume that an individual is infected with HIV first before getting exposed to Mtb infection. Then $\hat{T}^*$ the value of health CD4+ T cells at the time Mtb infection will occur is less than $\hat{T}$ the value of health CD4+ T cells when HIV infection occurs. This means that $\hat{T} > \hat{T}^*$. Now substituting $\hat{T}^*$ in the expression of R_{Tb_c} we get $R_{Tb_{\hat{T}^*_c}}$, $R_{Tb_c} < R_{Tb_{\hat{T}^*_c}}$, since $(k_b + k_a\frac{\hat{T}^*}{A} + k_l\hat{C}_b + \mu_{mi}) < (k_b + k_a\frac{\hat{T}}{A} + k_l\hat{C}_b + \mu_{mi})$. This suggests why there is fast progression of Mtb infection in an HIV co-infected individual.

4. Numerical Simulations

In this section, we study three models that is for Mtb and HIV separate infections when there is no co-infection, and the model for Mtb/HIV co-infection using numerical simulations. The C^{++} programming language was used to simulate the results in this section. In simulating co-infection results we started with a case (1) when Mtb and HIV infection occur at the same time or within a small window period, (2) when infection with one pathogen occurs first then infection with the second pathogen occurs after a considerable amount of time. The second scenario is complimented with a comparison of the severity of the two types of co-infection. The parameter values used are given in Table 2.

4.1. *Mtb infection*

Mtb infection occurs when an individual gets exposed to the Mtb pathogen and when his/her immune response fails to eradicate the pathogen. The infection will

Table 2. Table of parameters used in the model.

Name	Value	Definition	Units	Reference(s)
s_m	5	M_r source	M_r mm^{-3} day	12
p_m	0.000575	M_r proliferation	day^{-1}	Est
r_o	0.1	M_r HIV induced proliferation	day^{-1}	Est
k_v	0.000025	M_r rate of infection by HIV	day^{-1}	32
μ_m	0.011	M_r death rate	day^{-1}	22
k_b	0.00002	M_{ib} burst rate	day^{-1}	22
k_a	0.000125	Apoptosis	day^{-1}	22
A	350.0	CD4+ T Sat Limit	CD4+ T mm^{-3}	32
μ_{mi}	0.011	Death rate	day^{-1}	22
h_m	0.000045	Lytic killing of M_{iv}	mm^{-3}day^{-1}	Est
m_b	0.015	M_{iv} burst rate	day^{-1}	Est
k_l	0.005	Lysis of M_{ib}	mm^{-3}day^{-1}	15, 22
N_b	50.0	M_{ib} burst size	$T_b M_{ib}^{-1}$	15, 22
s_t	10.0	CD4+T supply rate	mm^{-3}day^{-1}	32
γ_1	0.00065	T_b killing by M_r	mm^{-3}day^{-1}	22
B	500.0	CD4+ Sat Limit due to T_b	CD4+ T mm^{-3}	Est
γ_2	0.02	T_b killing by C_b	mm^{-3}day^{-1}	Est
Aa	350.0	CD4+ Sat Limit due to HIV	CD4+ T mm^{-3}	32
r_m	0.1	T_b Multiplication	mm^{-3}day^{-1}	12, 15, 22
r_1	0.00001	CD4+ T Proliferation due to T_b	mm^{-3}day^{-1}	1
a_1	0.075	Infection inhibition factor	Scalar	Est
β	0.00025	HIV CD4+ T infection rate	mm^{-3}day^{-1}	32
R	350.0	Apoptosis Sat Limit	CD4+ T mm^{-3}day^{-1}	32
μ_t	0.02	CD4+ T death rate	day^{-1}	32
h_v	0.0025	Lysis of T^*	mm^{-3}day^{-1}	Est
s_v	5.0	CTL supply rate	mm^{-3}day^{-1}	32
r_2	0.00001	Apoptosis	mm^{-3}day^{-1}	Est
μ_v	1.5	HIV death rate	day^{-1}	32
α_t	0.025	T^* death rate	day^{-1}	32
p_v	0.00001	CTL proliferation	day^{-1}	32
s_b	5.0	Mtb CTL supply rate	mm^{-3}day^{-1}	Est
p_b	0.000001	Mtb CTL proliferation	mm^{-3}day^{-1}	22
μ_{cv}	1.5	HIV CTL death rate	day^{-1}	32
a_3	0.85	Virion production reduction factor	Scalar	Est
μ_{cb}	0.95	Mtb CTL death rate	day^{-1}	22
a_2	0.05	Virion production reduction factor	Scalar	Est
R_b	500.0	CTL proliferation Sat Limit	mm^{-3}day^{-1}	Est
N_V	1000.0	T^* virus burst size	VT^{*-1}	32
N_m	800.0	M_{iv} virus burst size	VM_{iv}^{-1}	Est

Est means estimated.

result in either latent TB or active TB depending on the robustness of the individuals immune response.

4.1.1. *Mtb latency*

TB latency occurs when the immune system is strong to the extend that it can prevent excessive bacterial multiplication that may cause active disease yet not strong enough to completely eradicate the pathogen from the body. It is normally evidenced by low extracellular bacteria and high intracellular bacteria.

Infected macrophages offer a protected environment that is favorable for bacterial multiplication. Our numerical simulations show that low burst rate of Mtb infected macrophages (k_b) and rate of extracellular bacterial multiplication coupled with high values of γ_1 (bacterial killing by resting macrophages), γ_2 (killing of bacteria by CTLs), and k_l (CTL lytic effect on Mtb infected macrophages) result in latency.

Figure 1a shows the decline in resting macrophages as a result of Mtb infection. Resting macrophages reach a constant level of about 260 macrophages per mm^3 after about 1000 days. The decrease in resting macrophages is matched with an increase in Mtb infected macrophages and bacterial load as shown in Figs. 1b and 1c, respectively. These results show that intracellular bacterial load is higher ($\approx 1400\,\text{mm}^{-3}$) than extracellular bacterial load ($\approx 640\,\text{mm}^{-3}$) and this is a marker of latency. This is a point of concern because once this balance is lost, maybe due to the weakening of the immune system as a result of co-infection, poor diet, aging and socio-economic factors then latency is lost (reactivation occurs) and extracellular bacteria load will bulge. Figures 1d and 1e show increase in T cells as Mtb infection progresses.

4.1.2. *Mtb disease*

Active TB disease occurs when there is high extracellular bacterial multiplication or high extracellular bacterial burden than intracellular bacterial load. We used higher values of bursting and extracellular bacterial multiplication rates and lower rates of immune response mechanisms that is, lower value of k_l, γ_1, and γ_2, than the values that we used when we simulated TB latency. When there is active TB, high bacterial burden and multiplication and relatively low intracellular bacterial load is noticed.

Simulation results in Figs. 2a and 2b show more pronounced decline of resting macrophages and increase in Mtb infected macrophages than noticed in Figs. 1a and 1b, respectively. Figure 2c shows rapid bacterial growth, which is a marker of active disease. Figures 2a and 2c show increase in T cells but not strong enough to control bacterial growth, hence occurrence of active TB.

4.2. *HIV infection*

HIV infection results in depletion of health CD4+ T cells. This weakens the immune system and in-turn makes the HIV patient vulnerable to opportunistic infections. In our simulations we show HIV infection that is in latency and that is also slowly approaching the AIDS stage. Simulation results not shown here reveal that infection of macrophages alone by the virus without infection of CD4+ T cells will not result in AIDS, but that a certain level of viral load is maintained and the immune system cannot eliminate it. Infection of macrophages alone is not a problem and studies[18,19] have shown that the strain that predominantly infects macrophages (CCR5 strain) can also infect CD4+ T cells and it also tends to mutate to the CXCR4 strain that predominantly infect CD4+ T cells. The CXCR4 strain is associated with fast

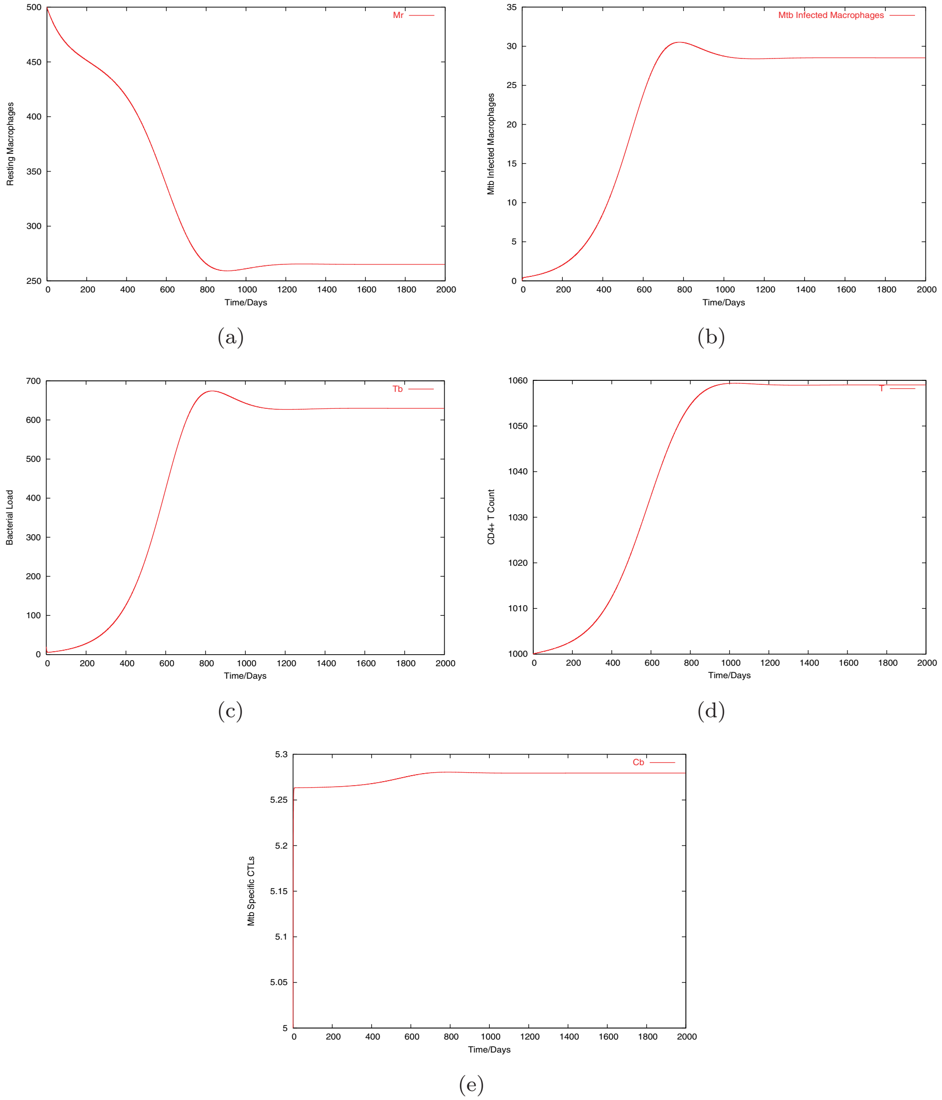


Fig. 1. Graphs of numerical solutions showing propagation of macrophages, T cells and Mtb pathogen during the first 2000 days of Mtb infection that results in TB latency: (a) resting macrophages, (b) Mtb infected macrophages, (c) Mtb pathogen, (d) CD4+ helper T cells, and (e) Mtb specific CTLs. Initial conditions: $M_r = 500.0$, $M_{ib} = 0.0$, $T_b = 20.0$, $T = 1000.0$, $C_b = 5.0$. Parameter values used to simulate latency: $k_i = 0.00002$, $k_b = 0.00001$, $k_l = 0.02$, $\gamma_2 = 0.02$, $\gamma_1 = 0.00065$, $r_m = 0.0625$.

progression of HIV infection to the AIDS stage. Our numerical simulations show that infection of both macrophages and CD4+ T cells aggravates the development of AIDS.

Figure 3a shows depletion of CD4+ T cells and Fig. 3b shows the increase in HIV infected CD4+ T cells. This depletion of health CD4+ T cells and increase

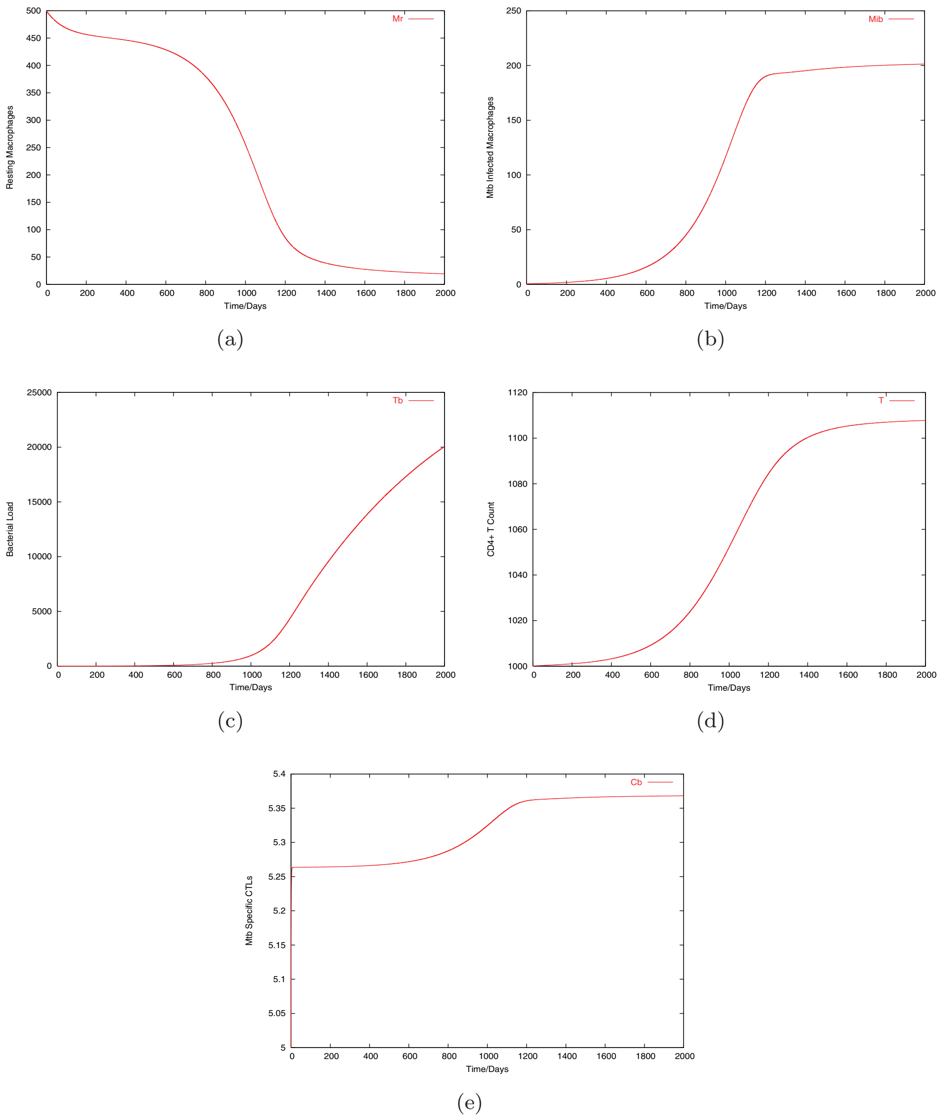


Fig. 2. Graphs of numerical solutions showing propagation of macrophages, T cells and Mtb pathogen during the first 2000 days of Mtb infection that results in TB disease: (a) resting macrophages, (b) Mtb infected macrophages, (c) Mtb pathogen, (d) CD4+ helper T cells, and (e) Mtb specific CTLs. Initial conditions: $M_r = 500.0$, $M_{ib} = 0.0$, $T_b = 20.0$, $T = 1000.0$, $C_b = 5.0$. Parameter values used to simulate TB disease: $k_i = 0.00002$, $k_b = 0.00002$, $k_l = 0.005$, $\gamma_2 = 0.01885$, $\gamma_1 = 0.00054$, $r_m = 0.1$.

in infected CD4+ T cells result in an increase of HIV virions that is noticed in Fig. 3d. As a result of depletion of CD4+ T cells, HIV specific CTLs tend to decline as noticed in Fig. 3c since CD4+ T cells are necessary for proliferation of HIV specific CTLs.

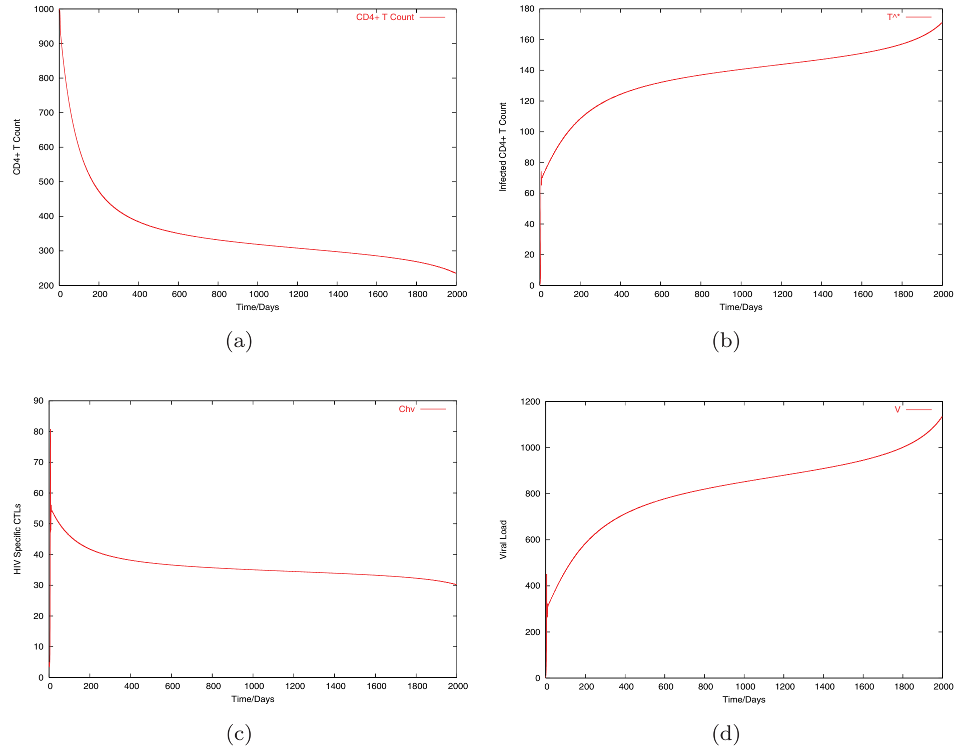


Fig. 3. Graphs of numerical solutions showing propagation of T cells and virus during the first 2000 days of HIV infection: (a) health CD4+ T cells, (b) infected CD4+ T cells, (c) HIV specific CTLs, and (d) HIV virions. Initial conditions: $T = 1000.0$, $T^* = 0.0$, $C_{hv} = 5.0$, $V = 10.0$.

4.2.1. *Co-infection*

Co-infection occurs when an individual gets infected with two different pathogens that cause separate diseases at the same time. This might arise from different scenarios: (1) co-infection can occur theoretically (simultaneously) at the same time or within a small window period, (2) infection can occur from one pathogen first then the same individual gets exposed to the second pathogen after a considerable time interval. Here we present simulations of co-infection with an assumption that infection occurs simultaneously or within a small window period, or co-infection with HIV can occur in an individual who is already latently infected with TB or an Mtb co-infection can occur as a result of exposing an HIV patient to Mtb infection. Our simulations show that due to co-infection latent TB is activated and an Mtb infection that resulted in active TB (Fig. 2) will progress to active disease even faster and HIV infection progression to the AIDS stage is enhanced.

Figure 4a shows the decline of resting macrophages during co-infection due to both Mtb and HIV pathogen. This decline gives rise both to Mtb and HIV infected macrophages shown in Figs. 4b and 4c, respectively. Figure 4 shows rapid bacterial

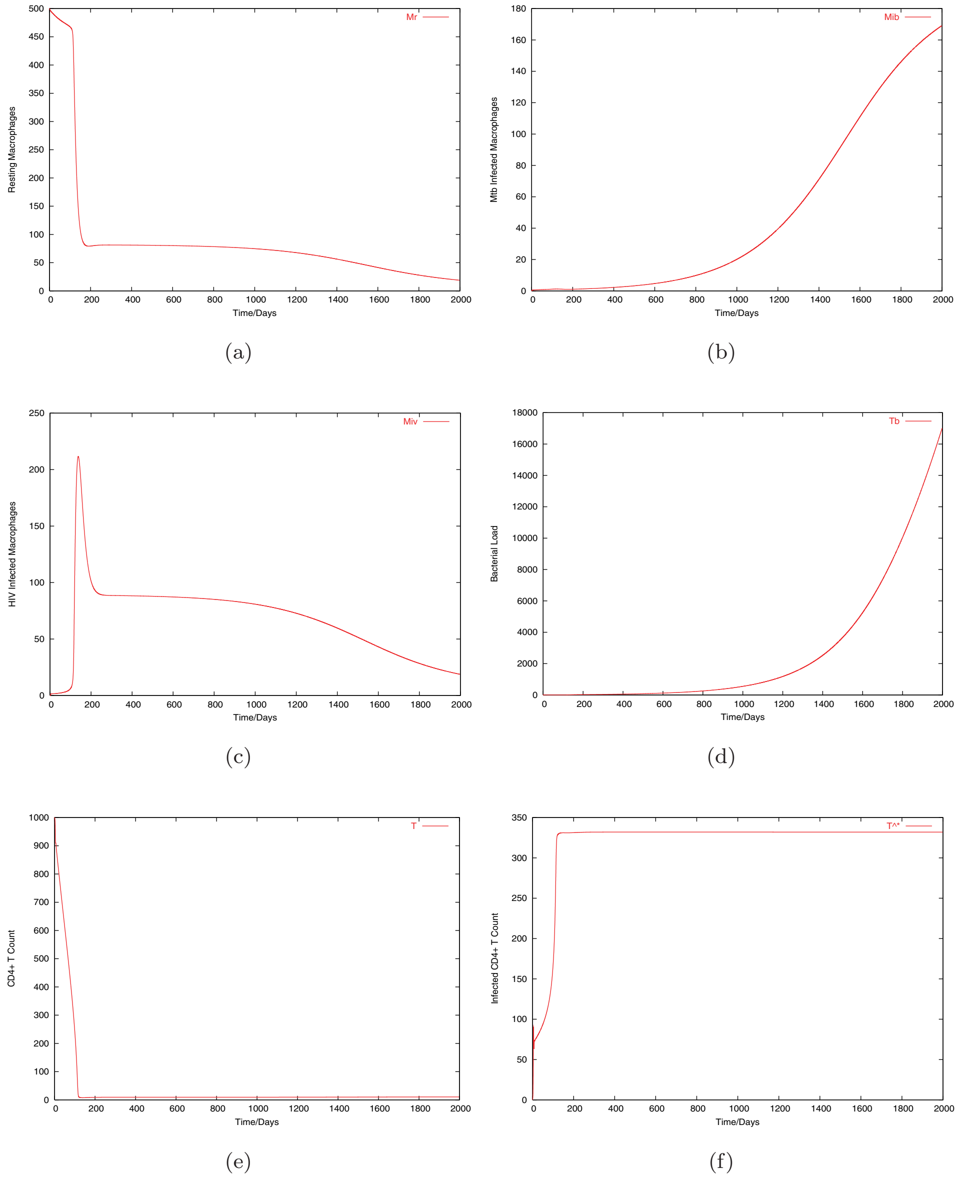
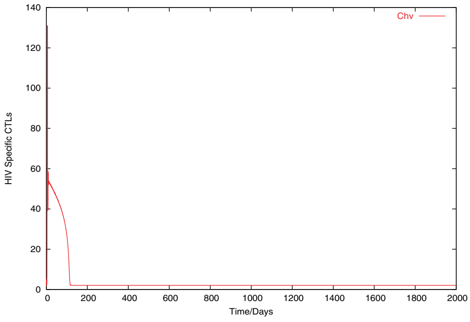
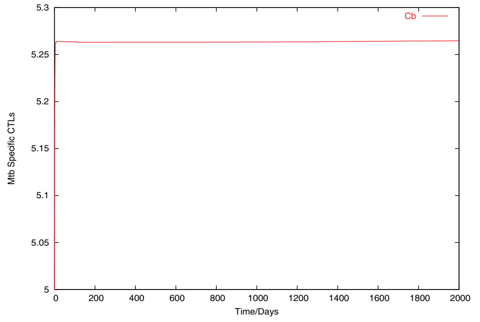


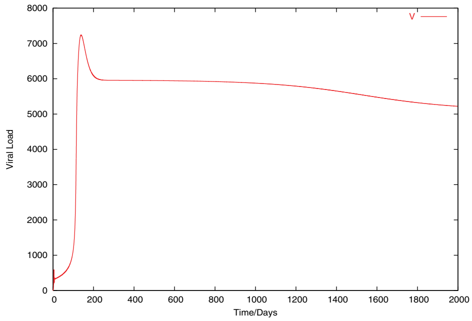
Fig. 4. Graphs of numerical solutions showing propagation of macrophages, bacteria, T cells, and virus during the first 2000 days of Mtb and HIV co-infection: (a) resting macrophages, (b) Mtb infected macrophages, (c) HIV infected macrophages, (d) Mtb pathogen, (e) health CD4+ T cells, (f) infected CD4+ T cells, (g) HIV specific CTLs, (h) Mtb specific CTLs, and (i) HIV virions. Initial conditions: $M_r = 500.0$, $M_{ib} = 0.0$, $M_{iv} = 0.0$, $T_b = 20.0$, $T = 1000.0$, $T^* = 0.0$, $C_{hv} = 5.0$, $C_b = 5.0$, $V = 10.0$.



(g)



(h)



(i)

Fig. 4. (*Continued*)

Table 3. Key used in Fig. 5 to distinguish how co-infection occurred.

Key	When co-infection occurs
Coinf-1	Mtb infection occurs when HIV is already slowly progressing to AIDS
Coinf-2	HIV infection when Mtb infection is already in latency

growth that occurs during co-infection. Figures 4e and 4f show depletion of health CD4+ T cells and increase of HIV infected CD4+ T cells. In Figs. 4g and 4h the dynamics of HIV specific CTLs and Mtb specific CTLs are shown. Viral growth is shown in Fig. 4i.

Figure 5 shows what happens when an individual who is suffering from infection with one of the pathogens get infected with the second pathogen. Figure 5a shows the changes in dynamics of resting macrophages. This graph shows that co-infection that starts with HIV infection results in a more rapid decline of resting macrophages and highest depletion of resting macrophages occurs from co-infection that starts from Mtb latency. Mtb infected macrophages also rise sharply in the case of Mtb latency when HIV co-infection occurs. Figure 5 shows activation of latency that

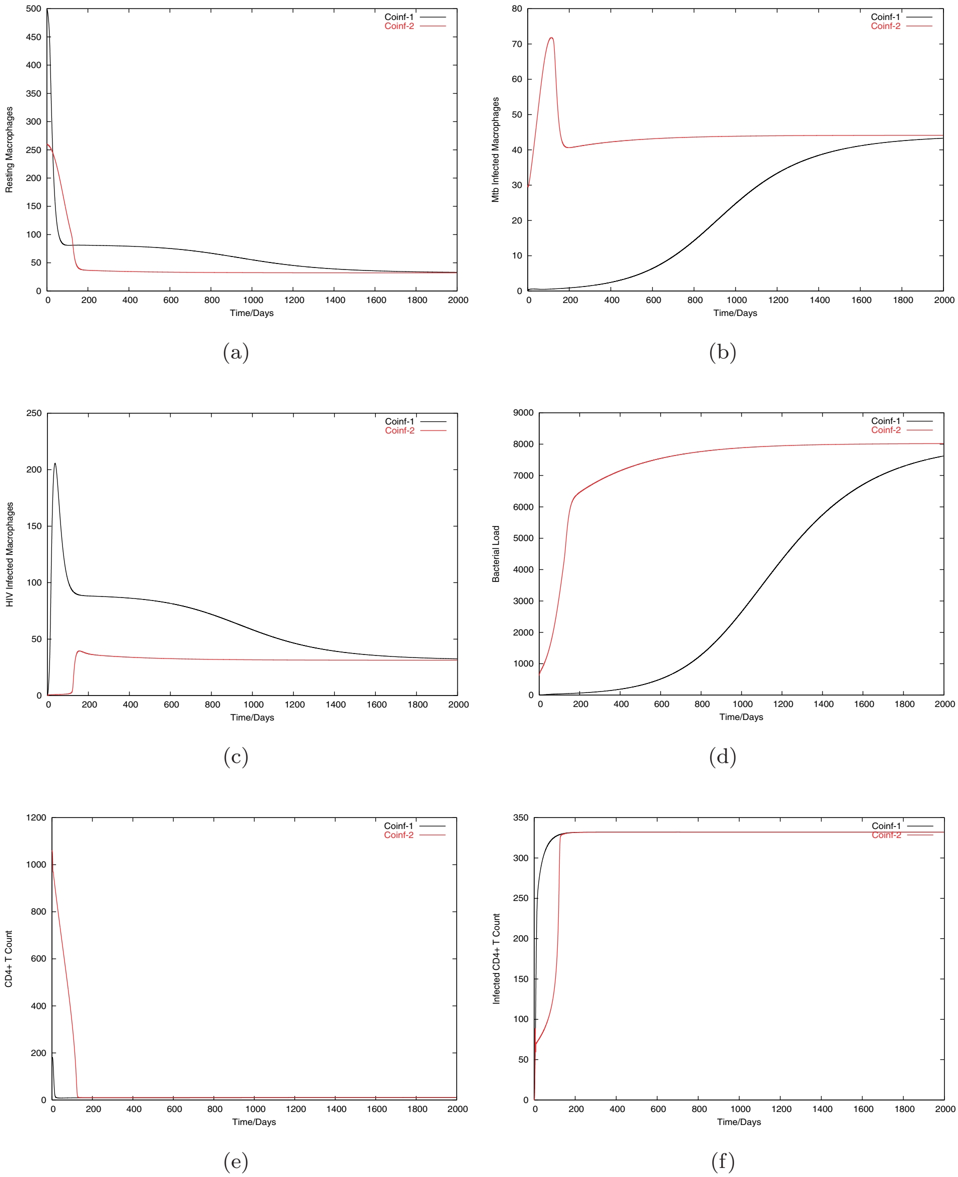
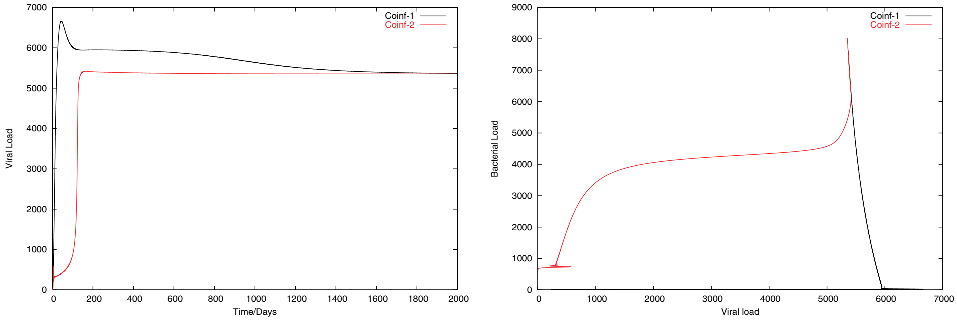


Fig. 5. Graphs of numerical solutions showing propagation of macrophages, bacteria, T cells, and virus that starts either from Mtb latency or HIV infection: (a) resting macrophages, (b) Mtb infected macrophages, (c) HIV infected macrophages, (d) Mtb pathogen, (e) health CD4+ T cells, (f) infected CD4+ T cells, (g) HIV virions, and (h) relationship of Mtb against HIV. Initial conditions: (i) Coinf-1, $M_r = 500.0$, $M_{ib} = 0.0$, $M_{iv} = 0.0$, $T_b = 20.0$, $T = 200.0$, $T^* = 170.0$, $C_{hv} = 30.0$, $C_b = 5.0$, $V = 1200.0$ and (ii) Coinf-2, $M_r = 260.0$, $M_{ib} = 30.0$, $M_{iv} = 0.0$, $T_b = 630.0$, $T = 1060.0$, $T^* = 0.0$, $C_{hv} = 5.0$, $C_b = 6.0$, $V = 10.0$.



(g) (h)

Fig. 5. (*Continued*)

we established in Fig. 1. When Mtb infection occurs in an HIV infected individual, Mtb infected macrophages tend to rise relatively slow, but proceed to active disease unlike in the case when there is no HIV co-infection as was shown in Fig. 1. The dynamics of Mtb infected macrophages are matched with the dynamical changes of extracellular bacteria in Fig. 5d. This shows activation of extracellular bacteria that were under suppression of the immune system when there was no co-infection. This graph also shows that extracellular bacteria will not be suppressed by the immune system as we showed in Fig. 2c when there is no HIV infection which weakens the immune system. We note that bacterial growth is highest when co-infection occurs starting from Mtb latency than Mtb co-infection that occurs starting in HIV infection. Macrophages are infected least by the virus when HIV co-infection occurs starting in an Mtb infection that is already in latency compared to the case when Mtb co-infection occurs starting from HIV infection. This corresponds to high viral load and infected CD4+ T cells that are noticed in Figs. 5g and 5f, respectively. This is associated with the assumption that we made, that infection of macrophages in the lungs by the virus occurs only when there is co-infection. The low level of HIV infected macrophages corresponds to low viral load and a delayed increase in infected CD4+ T cells that is noticed in Figs. 5g and 5f, respectively. Figure 5e shows that high viral load result in rapid CD4+ T cell depletion. Figure 5h shows the relationship between Mtb and HIV that arises from the two different scenarios of co-infection. When co-infection with HIV occurs in an already latently Mtb infected state, bacterial load grow instantly at a fast rate while if an Mtb co-infection occurs in an already HIV infected state; HIV infection multiply at a faster rate while Mtb growth maintain a low profile until a certain level of viral load is reached then Mtb density grows rapidly. The graphs merge after a considerable time of about 3 to 4 years as shown in Figs. 5a–5d and 5g.

Generally these simulations show that co-infection aggravates both infections irrespective of which infection started but more importantly to take note of is the fact that the levels of aggravation differs depending on which infection started.

Mtb infection is aggravated more by HIV co-infection when we start in Mtb latency than when we get an Mtb co-infection starting from an HIV infection. The same is also true for HIV infection. HIV infection is aggravated the most when Mtb co-infection occurs in an already HIV infected individual than when we start in Mtb infection. From this observation we conclude that co-infection favors fast progression of the pathogen that caused the first infection and from this we suggest that careful/timed treatment of the first disease inducing pathogen must be administered. Also our numerics show that this difference in disease progression patterns vanishes as co-infection progresses since the patterns tend to converge to one trend after about 3 to 4 years of co-infection. This suggests that delaying administration of treatment is not favorable since there will be no differences or not be possible to separate or determine whether the infection started initially from Mtb infection or from HIV infection.

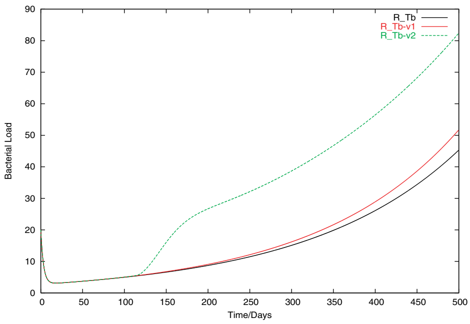
4.2.2. R_{Tb} *and* R_{HIV} *analysis during co-infection*

Figure 6a shows the effect of co-infection to the Mtb disease progression. Disease progression tends to increase as a result of co-infection and even faster when there is both infection of macrophages by the virus and the Mtb pathogen. Also HIV progression is shown to be enhanced due to co-infection (Fig. 6b). Highest HIV infection progression occurs during co-infection when there is both infection of macrophages and T cells by the virus and is least when there is no co-infection. In Fig. 6c we show the relationship between Mtb infection against HIV infection. This figure shows that as viral load increases the bacterial load does not really change until the viral load has reached a certain level. When this level is reached then bacterial load erupts. This means that infection with HIV of an Mtb infected individual does not mean the individual will instantly develop active TB, but active TB will only occur after and until HIV has depleted or weakened the immune system to a certain level. Once the immune system has been weakened to this level the individual will become vulnerable to any opportunistic infection.

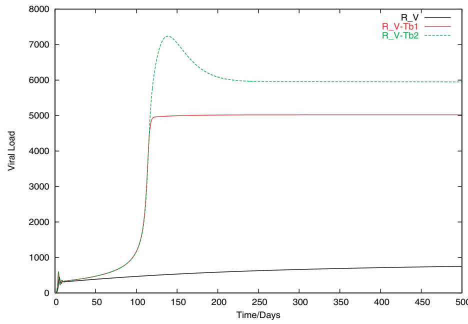
Figure 7 shows the effect of co-infection on the depletion of resting macrophages by the Mtb pathogen and CD4+ T cells by HIV. Highest depletion of these cells occurs during co-infection when there is both infection of macrophages by the virus and by the Mtb pathogen and is lowest when there is no co-infection.

5. Discussion

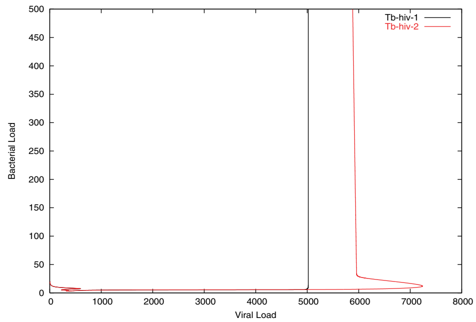
This work shows the significance of CD4+ T cells and specific CTL mechanisms in the control of Mtb infection, HIV infection and during TB/HIV co-infection. We hypothesized that the ratio of Mtb CTLs to HIV CTLs is proportional to the ratio of Mtb bacteria to HIV viral load in the lungs and that Mtb/HIV CTLs are inversely proportional to Mtb/HIV pathogen levels in the lungs. The work of



(a)

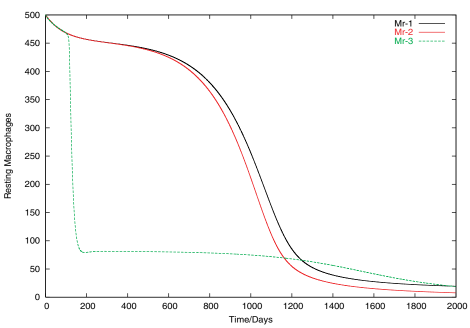


(b)

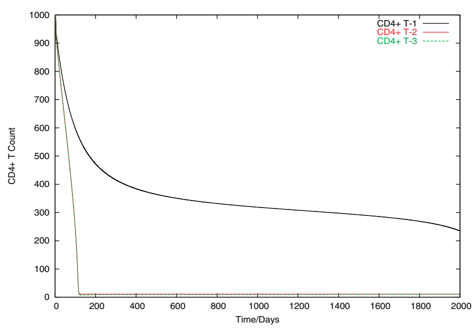


(c)

Fig. 6. Graphs of numerical solutions showing changes in propagation of Mtb and HIV infection progression patterns due to co-infection and the relationship between Mtb and HIV infections during co-infection: (a) change in Mtb kinetics as a result of co-infection, (b) change in HIV infection progression rate due to co-infection, and (c) the relationship between Mtb and HIV infection during co-infection.



(a)



(b)

Fig. 7. Graphs of numerical solutions showing changes in propagation of macrophages and CD4+ T during co-infection: (a) resting macrophages population changes when there is and when there is no co-infection and (b) depletion of health CD4+ T cells during no and during co-infection.

Kirschner[1] illustrated the combined effects of CD4+ T cells and CD8+ T cells during co-infection in a model that sort to determine the impact of TB on HIV progression in the lypmh tissue and the attributes of separate CD4+ T cells, Mtb CTLs and HIV CTLs are missing in their studies.[1] Since TB is a disease of the lungs we considered the co-infection dynamics of TB and HIV in the lungs and how each infection worsen the picture of the other. HIV infection results in the depletion of both macrophages and CD4+ T cells and these cells are important in the immune mechanism required to control Mtb infection. Infection of macrophages by HIV show that apart from infected CD4+ T cells, macrophages are a potential reservoir of viral particles. Mtb infection also has the capacity to induce rapid replication of HIV through production of cytokines like TNF-α[2–4] this results in high viral load, hence more depletion of CD4+ T cells. As a result of TB/HIV co-infection our results show that the progression of Mtb infection is enhanced. Our study further shows how co-infection enhance infection of macrophages by the X4 viral strain in the lungs a phenomenon that rarely occurs when there is no co-infection. Our numerical and analytical analysis demonstrate that there is increased viral production in the lungs from two source (infected macrophages and CD4+ T cells) that weakens the immune mechanisms that are necessary for the control of Mtb infection. On the other hand infection of macrophages by the virus offers HIV a protected environment to multiply since they are generally resistant to the cytopathic mechanisms of the virus. This in-turn favors the progression of HIV infection during co-infection. It is necessary to point out that infection of macrophages is also noticed even in the absence of co-infection. But a point to take home is that this infection is mainly by the R5 viral (that are M-tropic) strain and little or no virus multiplication is noticed in the lungs even in advanced AIDS patients.[18,19] Then TB infection in HIV infects expand the co-receptor usage and up-regulate the CXCR4 coreceptors expression on macrophages this favors preferential production of the X4 viral strain and significant infection of macrophages in the lungs. As a result Mtb infection accelerates the course of HIV and in-turn HIV also worsen the picture to Mtb infection. Another interesting point to take note of is the increased rate of depletion of the immune cells that our numerical simulations portray as a result of the co-infection. This points to the fast progression of both infections. Our numerical simulations also show that co-infection accelerates more the first pathogen that induced infection before co-infection occurred. This suggests special treatment or timed treatment of the first pathogen that induced the first infection before co-infection occurred as a result of the second disease inducing pathogen or timed treatment of the two pathogen as soon as co-infection is discovered. Early treatment of the first disease inducing pathogen when co-infection is discovered will help to reduce exacerbation of the second pathogen causing co-infection. If treatment is delayed then the second pathogen inducing co-infection is aggravated to progress fast such that after a certain time interval of co-infection it will not be easy to distinguish the pathogen that caused co-infection, since the disease progression patterns tend to converge.

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References

1. Kirschner D, Dynamics of co-infection with *M. tuberculosis* and HIV-1, *Theor Popul Biol* **55**:94–109, 1999.
2. Weiden M *et al.*, Differentiation of monocytes to macrophages switches the *Mycobacterium tuberculosis* effect on HIV-1 replication from stimulation to inhibition: modulation of interferon response and CCAAT/enhancer binding protein β expression, *J Immunol* **165**:2028–2039, 2000.
3. Honda Y *et al.*, Type I interferon induces inhibitory 16-kD CCAAT/enhancer binding protein (C/EBP)β, repressing the HIV-1 long terminal repeat in macrophages: pulmonary tuberculosis alters C/EBP expression, enhancing HIV-1 replication, *J Exp Med* **188**(7):1255–1265, 1998.
4. Kaufmanns SH and Schaible UE, A dangerous liaison between two major killers: *Mycobacterium tuberculosis* and HIV target dendritic cells through DC-SIGN. Commentry, *J Exp Med* **197**:121–127, 7–17, 2003.
5. Kirschner D, Using mathematics to understanding HIV immune dynamics, *Not AIMS* **43**:191–202, 1996.
6. Janeway AC, Travers P, Walport M, Shlomchik JM, *Immuno Biology: The Immune System in Health and Disease*, Garland Science Publishing, 2005.
7. Stevenson M, HIV-1 pathogenesis: review, *Nat Med* **9**(7):853–860, 2003.
8. Kaplan D, Sieg S, Role of the Fas/Fas ligand apoptotic pathway in human immunodeficiency virus type 1 disease, *J Virol* **72**(8):6279–6282, 1998.
9. Raja A, Immunology of tuberculosis: review article, *Indian J Med Res* **120**:213–232, 2004.
10. Schluger NW, Rom WN, The host immune response to tuberculosis: state of the art, *Am J Respir Crit Care Med* **157**:679–691, 1998.
11. Flynn JL, Immunology of tuberculosis and implications in vaccine developments, *Tuberculosis* **84**:93–101, 2004.
12. Wigginton JE, Kischner D, A model to predict cell mediated immune regulatory mechanisms during human infection with *Mycobacterium tuberculosis*, *J Immunol* **166**(3):1951–1967, 2001.
13. Schluger NW, Recent advances in our understanding of the human host responses to tuberculosis: review, *Respir Res* **2**:157–163, 2001.
14. Rook GAW, Seah G, Ustianowski A, *M. tuberculosis*: immunology and vaccination, *Eur Respir* **17**:537–557, 2001.
15. Marino S, Kirschner D, The human immune response to *Mycobacterium tuberculosis* in lung and lymph node, *J Theor Biol* **227**:463–486, 2004.
16. Stenger S *et al.*, An anti-microbial activity of cytotoxic T cells mediated by granulysin, *Science* **282**:121–125, 1998.
17. Stenger S *et al.*, Differential effects of cytolytic T cell subsets on intracellular infection, *Science* **276**:1684–1687, 1997.
18. Hoshino Y *et al.*, Maximal HIV-1 replication in alveolar macrophages during tuberculosis requires both lymphocyte contact and cytokines, *J Exp Med* **195**(4):495–505, 2002.

19. Hoshino Y *et al.*, *Mycobacterium tuberculosis*-induced CXR4 and chemokine expression leads to preferential X4 HIV-1replication in human macrophages, *J Immunol* **172**:6251–6258, 2004.
20. Yi Y *et al.*, Role of CXCR4 in cell-cell fusion and infection of monocyte derived macrophages by primary human immunodeficiency virus type 1 (HIV-1) strains: two distinct mechanisms of HIV-1 dual tropism, *J Virol* **73**(9):7117–7125, 1999.
21. Raju B *et al.*, *In situ* activation of helper T cells in the lung, *Infect Immun* **69**(8): 4790–4798, 2001.
22. Magombedze G, Garira W, Mwenje E, Modelling the human immune response mechanisms to *Mycobacterium tuberculosis* infection in the lungs, *MBE* **3**(4):661–682, 2006.
23. Flynn JL, Goldstein MM, Chan J, Triebold KJ, Pfeffer K, Lowenstein CJ *et al.*, Tumor necrosis factor-a is required in the protective immune response against *Mycobacterium tuberculosis* in mice, *Immunity* **2**:561–572, 1995.
24. Diekmann O *et al.*, On the definition and computation of the basic reproduction R_0 in models for infectious diseases in heterogeneous population, *J Math Biol* **28**:365–382, 1990.
25. Castillo–Chavez C *et al.*, On the computation R_0 its role on global stability, *IMA* **125**:229–250, 2002.
26. Rosas Taraco AG *et al.*, *Mycobacterium tuberculosis* upregulates coreceptors CCR5 and CXCR4 while HIV modulates CD14 favoring concurrent infection, *AIDS Res Hum Retrovir* **22**(1):45–51, 2006.
27. Aquaro S *et al.*, Macrophages and HIV infection: therapeutical approaches toward this strategic virus reservoir, *Antiviral Res* **55**:209–225, 2002.
28. Moriuchi M, Moriuchi H, Turner W, Fauci AC, Exposure to bacterial products renders macrophages highly susceptible to T-tropic HIV-1, *J Clin Invest* **102**(8):1540–1550, 1998.
29. Orenstein JN, The macrophage in HIV infection, *Immunobiology* **204**:598–602, 2001.
30. Verani A, Gras G, Pancino, Macrophages and HIV-1: dangerous liaisons, *Mol Immunol* **42**:195–212, 2005.
31. Serbina NV, Flynn JL, Early emergence of CD8+ T cells primed for production of Type 1 cytokenes in the lungs of *Mycobacterium tuberculosis*-infected mice, *Infect Immunol* **67**:3980–3988, 1999.
32. Shiri T, Garira W, Musekwa SD, A two-strain HIV-1 mathematical model to assess the effects of chemotherapy on disease parameters, *MBE* **2**:811–832, 2005.

Appendix A

Proof of Theorem 1. $F(X_G, 0)$, A_G $(= D_{Z_G}G(X_G^*, 0))$, and $\hat{G}(X_G, Z_G)$ and are given as follows:

$$\mathbf{F}(\mathbf{X_G}, \mathbf{0}) = \begin{pmatrix} s_m - \mu_m M_r \\ s_t - \mu_t T \\ s_b - \mu_{cb} C_b \end{pmatrix},$$

$$\mathbf{A_G} = \begin{pmatrix} -\left(k_b + k_a \dfrac{\hat{T}}{A} + k_l \hat{C}_b + \mu_{mi}\right) & k_i \hat{M}_r \\ k_b N_b + k_l N_c \hat{C}_b & -(\gamma_1 \hat{M}_r + \gamma_2 \hat{C}_b - r_m) \end{pmatrix},$$

$$\hat{\mathbf{G}}(\mathbf{X_G},\mathbf{Z_G})=\begin{pmatrix} M_{ib}k_a(\frac{T}{M_{ib}+A}-\frac{\hat{T}}{A})+M_{ib}k_l(C_b-\hat{C}_b)+T_bk_i(\hat{M}_r-M_r) \\ M_{ib}k_lN_c(\hat{C}_b-C_b)+T_b\gamma_1(M_r-\hat{M}_r)+T_b\gamma_2(C_b-\hat{C}_b) \end{pmatrix}.$$

Global stability of the system at E_{00} requires that $\hat{G}(X_G,Z_G)\geq 0$. Note, (i) $T>\hat{T}$, since $B>A$, $\hat{T}=(\frac{s_t}{\mu_t})$ and $T=\frac{s_t}{\mu_t-(r_1\frac{T_b}{T_b+B})}$, $\big(=(\frac{s_t}{\mu_t-r_1T_b})\big(\frac{T_b+B}{\frac{T_b}{N_b}+A}\big)\big)$, then it follows that $(\frac{T}{M_{ib}+A}>\frac{\hat{T}}{A})$. (ii) $C_b>\hat{C}_b$ when there is Mtb infection, since $\hat{C}_b=(\frac{s_b}{\mu_{cb}})$ and $C_b=(\frac{s_b}{\mu_{cb}-p_b(\frac{M_{ib}}{M_{ib}+R_b})T})$ and (iii) $\hat{M}_r>M_r$ if $k_i>p_m$, $M_r=\frac{s_m}{\mu_m+T_b(k_i-p_m)}$, $\hat{M}_r=\frac{s_m}{\mu_m}$. Therefore, $\hat{G}_1(X_G,Z_G)\geq 0$ if $k_i>p_m$. (iv) $\hat{G}_2(X_G,Z_G)\geq 0$, if $(T_b\gamma_2-M_{ib}k_lN_c)(C_b-\hat{C}_b)\geq T_b\gamma_1(M_r-\hat{M}_r)$, since $\gamma_2>\gamma_1>k_l$. It is easy to show that $X_G^*=(\frac{s_m}{\mu_m},\frac{s_t}{\mu_t},\frac{s_b}{\mu_{cb}})$ is globally asymptotically stable state of $\frac{dX_G}{dt}=F(X_G,0)$, since $F(X_G,0)$ is a limiting function of $\frac{dX_G}{dt}=F(X_G(t),Z_G(t))$, that is, $\lim_{t\to\infty}X_G(t)=X_G^*$. Therefore, if $\hat{G}_1(X_G,Z_G),\hat{G}_2(X_G,Z_G)\geq 0$ then E_{00} is globally stable. If this condition fails to hold, then E_{00} may not be globally asymptotically stable.

Appendix B

Proof of Theorem 2. $F_B(X_B,0)$, A_B $(=D_{Z_B}G_B(X_B^*,0))$, and $\hat{G}_B(X_B,Z_B)$ and are given as follows:

$$\mathbf{F_B}(\mathbf{X_B},\mathbf{0})=\begin{pmatrix} s_t-\mu_tT \\ s_v-\mu_{cv}C_{hv} \end{pmatrix},$$

$$\mathbf{A_B}=\begin{pmatrix} -(h_v\hat{C}_{hv}+\alpha_t) & \frac{\beta\hat{T}}{1+a_1\hat{C}_{hv}} \\ \frac{N_v\alpha_t}{1+a_2\hat{C}_{hv}} & -\mu_v \end{pmatrix},$$

$$\hat{\mathbf{G}}_{\mathbf{B}}(\mathbf{X_B},\mathbf{Z_B})=\begin{pmatrix} T^*h_v(C_{hv}-\hat{C}_{hv})+\beta V\left(\frac{\hat{T}}{1+a_1\hat{C}_{hv}}-\frac{T}{1+a_1C_{hv}}\right) \\ N_v\alpha_tT^*\left(\frac{1}{1+a_2\hat{C}_{hv}}-\frac{1}{1+a_2C_{hv}}\right) \end{pmatrix}.$$

$\hat{G}_{1B}(X_B,Z_B)\geq 0$, (i) since $C_{hv}>\hat{C}_{hv}$, $(C_{hv}=\frac{s_v}{\mu_{cv}-p_vVT}$, $\hat{C}_{hv}=\frac{s_v}{\mu_{cv}}$ and $p_vVT>0$ always when there is HIV infection). (ii) $\hat{T}>T$, $(\hat{T}=\frac{s_t}{\mu_t})$, but the endemic equilibrium value of T is difficult to evaluate. Nevertheless biological facts and our numerical simulation estimates prove that depletion of CD4+ T cells occur. If $C_{hv}>\hat{C}_{hv}$, then it follows that $\frac{1}{1+a_{1,2}\hat{C}_{hv}}>\frac{1}{1+a_{1,2}C_{hv}}$. Therefore, $\hat{G}_{2B}(X_B,Z_B)$ is also greater than or equal to zero. The global stability of $X_B^*=(\frac{s_t}{\mu_t},\frac{s_v}{\mu_{cv}})$ is easy to verify. This analysis shows that $\hat{G}_B(X_B,Z_B)\geq 0$, therefore E_{10} is globally asymptotically stable.

Appendix C

Proof of Theorem 3. $F_C(X_C, 0)$, A_C $(= D_{Z_C} G_C(X_C^*, 0))$, and $\hat{G}_C(X_C, Z_C)$ and are given as follows:

$$\mathbf{F_C}(\mathbf{X_C}, \mathbf{0}) = \begin{pmatrix} s_m - \mu_m M_r \\ s_t - \mu_t T \\ s_v - \mu_{cv} C_{hv} \\ s_b - \mu_{cb} C_b \end{pmatrix},$$

$$\mathbf{A_C} = \begin{pmatrix} -k_b - \frac{k_a \hat{T}}{A} - k_l \hat{C}_b - \mu_{mi} & 0 & 0 & k_i \hat{M}_r & 0 \\ 0 & -h_m \hat{C}_{hv} - m_b & 0 & 0 & \frac{k_v \hat{M}_r}{1+a_o \hat{C}_{hv}} \\ 0 & 0 & -h_v \hat{C}_{hv} - \alpha_t & 0 & \frac{\beta \hat{T}}{1+a_1 \hat{C}_{hv}} \\ (k_b N_b + k_l N_c \hat{C}_b) & 0 & 0 & -\gamma_1 \hat{M}_r - \gamma_2 \hat{C}_b + r_m & 0 \\ 0 & \left(\frac{N_m m_b}{1+a_3 \hat{C}_{hv}}\right) & \left(\frac{N_v \alpha_t}{1+a_2 \hat{C}_{hv}}\right) & 0 & -\mu_v \end{pmatrix},$$

$$\hat{\mathbf{G}}_\mathbf{C}(\mathbf{X_C}, \mathbf{Z_C}) = \begin{pmatrix} M_{ib} k_a \left(\frac{T}{M_{ib}+A} - \frac{\hat{T}}{A}\right) + k_l M_{ib}(C_b - \hat{C}_b) + T_b k_i (\hat{M}_r - M_r) \\ M_{iv} h_m (C_{hv} - \hat{C}_{hv}) + k_v \left(\frac{\hat{M}_r}{1+a_o \hat{C}_{hv}} - \frac{M_R}{1+a_o C_{hv}}\right) \\ T^* h_v (C_{hv} - \hat{C}_{hv}) + \beta \left(\frac{\hat{T}}{1+a_1 \hat{C}_{hv}} - \frac{T^*}{1+a_1 C_{hv}}\right) \\ k_l N_c M_{ib} (\hat{C}_b - C_b) + \gamma_1 T_b (M_r - \hat{M}_r) + \gamma_2 T_b (C_b - \hat{C}_b) \\ N_m m_b M_{iv} \left(\frac{1}{1+a_3 \hat{C}_{hv}} - \frac{1}{1+a_3 C_{hv}}\right) + N_v \alpha_t T^* \left(\frac{1}{1+a_2 \hat{C}_{hv}} - \frac{1}{1+a_2 C_{hv}}\right) \end{pmatrix}.$$

For global stability $\hat{G}_C(X_C, Z_C) \geq 0$, that is, $\hat{G}_{1C}(X_C, Z_C)$, $\hat{G}_{2C}(X_C, Z_C)$, $\hat{G}_{3C}(X_C, Z_C)$, $\hat{G}_{4C}(X_C, Z_C)$, $\hat{G}_{5C}(X_C, Z_C) \geq 0$. It is simple to show that $\hat{G}_{2C}(X_C, Z_C), \hat{G}_{3C}(X_C, Z_C)$, $\hat{G}_{5C}(X_C, Z_C) \geq 0$, since $\hat{T} > T$(due to the depletion of T cells by the virus), $C_{hv} > \hat{C}_{hv}$ (as we showed earlier in the global stability of HIV infection only), $\hat{M}_r > M_r$, $\left(\hat{M}_r = \frac{s_m}{\mu_m}, M_r = \frac{s_m}{\mu_m + T_b(k_i - p_m) + V(\frac{k_v}{1+a_o C_{hv}} - p_m r_o)}\right)$, if $k_i > p_m$, and $k_v > p_m r_o$. This can be established by comparing the parameter values used in the numerical simulations and by substituting the equilibrium values of the variables at the endemic equilibria. The analysis of $\hat{G}_{1C}(X_C, Z_C)$ and $\hat{G}_{4C}(X_C, Z_C)$ is similar to the analysis of $\hat{G}_{1G}(X_G, Z_G)$ and $\hat{G}_{2G}(X_G, Z_G)$ of the model of Mtb infection only in Appendix A, respectively. The global stability of $X_C^* = (\frac{s_m}{\mu_m}, \frac{s_t}{\mu_t}, \frac{s_v}{\mu_{cv}}, \frac{s_b}{\mu_{cb}})$ of the system $\frac{dX_C}{dt} = F(X_C, 0)$ is not difficult to check. Therefore, if $\hat{G}_C(X_C, Z_C) \geq 0$ E_{20} is asymptotically stable and the same conclusion follows as in the model of Mtb infection only.